

Jaypee Institute of Information Technology, Noida

**Department of
Electronics and Communication Engineering**



Basic Electronics Lab (24B15EC111)

Semester : Odd (Ist) Semester



JAYPEE INSTITUTE OF INFORMATION TECHNOLOGY, NOIDA

VISION

To become a centre of excellence in the field of IT and related emerging areas of education, training and research comparable to the best in the world for producing professionals who shall be leaders in innovation, entrepreneurship, creativity and management.

MISSION

- To develop as a benchmark university in emerging technologies.
- To provide state-of-the-art teaching learning process and R&D environment.
- To harness human capital for sustainable competitive edge and social relevance.

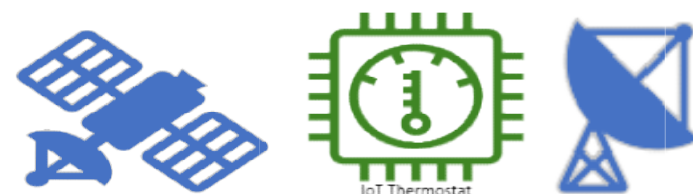
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

VISION

To be a centre of excellence in education, training and research in Electronics and Communication Engineering to cultivate technically competent professionals for Industry, Academia and Society.

MISSION

- To impart education through contemporary, futuristic and flexible curricula with innovative teaching learning methods and hands on training with well equipped Labs.
- To carry out cutting edge research in different areas of Electronics and Communication Engineering.
- To inculcate technical and entrepreneurial skills in professionals to provide socially relevant and sustainable solutions.



PROGRAMME : B.TECH. IN ELECTRONICS AND COMMUNICATION ENGINEERING

PROGRAMME EDUCATIONAL OBJECTIVES:

PEO1: To provide strong foundation in Electronics and Communication Engineering to pursue professional career, entrepreneurship and higher studies.

PEO2: To develop capability to analyze, design and develop feasible solutions to real world problems.

PEO3: To inculcate professional ethics, managerial and communication skills to develop ingenious solutions for benefit of society and environment.

PROGRAMME OUTCOMES:

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAMME SPECIFIC OUTCOMES:

PSO 1: To identify the engineering problems and develop solutions in the area of communication, signal processing, VLSI and embedded systems.

PSO2: To demonstrate proficiency in utilisation of software and hardware tools along with analytical skills to arrive at appropriate solutions.

Detailed Syllabus

Course Code	24B15EC111	Semester: Odd (specify Odd/Even)	Semester: 1st Session 2024-25 Month from: July to December
Course Name	Basic Electronics Lab		
Credits	1	Contact Hours	2

Faculty (Names)	Coordinator(s)	Samriti Kalia, Vinay Anand Tikkiwal
	Teacher(s) (Alphabetically)	Abhishek Kashyap , Abhay Kumar, Alok Joshi, Ankur Bhardwaj, Archana Pandey, Divya Kaushik, Garima Kapoor, Nitin Muchhal, Varun Goel, RituRaj, K. Nisha, Mandeep Narula, Satyendra Kumar, Shamim Akhtar, Yogesh Kumar

COURSE OUTCOMES - At the end of the course, students will be able to:		COGNITIVE LEVELS
C141.1	Recall various electronic components and working of basic measuring instruments	Remembering Level (C1)
C141.2	Understand the input-output characteristics of BJT	Understanding Level (C2)
C141.3	Verify Kirchhoff's laws and apply network theorems to solve DC circuit	Applying Level (C3)
C141.4	Analyze operational amplifier in various configurations and characteristics of basic diodes including their applications	Analyzing Level (C4)

Module No.	Title of the Module	List of Experiments	CO
1.	Introduction to basic electrical equipment and components	Introduction to various components (Resistor, Capacitor, Inductor, and IC) and instruments Multimeter, Bread board, Regulated D.C. power supply, and CRO.	C141.1
2.	Basic Circuit Analysis	Verification of KVL and KCL using a given circuit.	C141.3
3.	Basic Circuit Analysis	Verification of Thevenin's theorem.	C141.3
4.	PN Junction diode and Applications	To study the forward bias I-V (current-voltage) characteristics of a simple p-n junction diode. Also determine the forward resistance of the diode	C141.4
5.	PN Junction diode and Applications	To observe the output waveform of half/full wave rectifier and calculate its ripple factor and efficiency	C141.4
6.	Zener diode and Applications	To study the reverse bias I-V (current-voltage) characteristics of a Zener diode. Also determine the breakdown voltage, static and dynamic resistances.	C141.4
7.	Bipolar Junction Transistors	To plot input characteristics of a common emitter NPN BJT	C141.2

8.	Bipolar Junction Transistors	To plot output characteristics of a common emitter NPN BJT	C141.2
9	Operational Amplifier	To realize inverting and non inverting amplifier configuration using Op-Amp IC- 741	C141.4
10.	Operational Amplifier	To realize adder and subtractor circuits using Op-Amp IC-741	C141.4
11.	Basic Circuit Analysis	Verification of Superposition Theorem.	C141.3
12.	PN Junction diode and Applications	Realization of desired wave shapes using clipper and clamper circuits	C141.4
13.	Virtual Lab Experiments	To plot input characteristics of a common collector NPN BJT.	C141.2
14.	Virtual Lab Experiments	To plot output characteristics of a common collector NPN BJT.	C141.2

Evaluation Criteria

Components	Maximum Marks
Mid Sem Viva	20
End Sem Viva	20
Day-to-day performance, Lab Record	60
Total	100

Project Based Learning: Students will learn working of basic electronic equipment and applications of basic circuit theorems and different semiconductor devices including diodes and transistors to design circuits for various applications.

Recommended Reading material: Author(s), Title, Edition, Publisher, Year of Publication etc. (Text books, Reference Books, Journals, Reports, Websites etc. in the IEEE format)

1.	R. L. Boylestad, and L. Nashelsky, “Electronic Devices and Circuit Theory”, 11 th Ed., Prentice Hall of India, 2014.
2.	D.C. Kulshreshtha, “Basic Electrical Engineering”, Revised 1 st Ed., Tata McGraw Hill, 2017
3.	S.M. Sze, K.K. Ng, “Physics of Semiconductor Devices”, Wiley India, 3 rd Ed., 2006.
4.	R. A. Gayakwad, “Op-Amps and Linear Integrated Circuits”, 4 th Ed., Pearson, 2000.

Jaypee Institute of Information Technology, Noida

Department of ECE



List of Experiments

Basic Electronics Lab (24B15EC111) (Odd Semester)

1. Introduction to various components (Resistor, Capacitor, Inductor, and IC) and instruments Multimeter, Bread board, Regulated D.C. power supply, and CRO. (C141.1)
2. Verification of KVL and KCL using a given circuit. (C141.3)
3. Verification of Thevenin's theorem. (C141.3)
4. To study the forward bias I-V (current-voltage) characteristics of a simple p-n junction diode. Also determine the forward resistance of the diode. (C141.4)
5. To observe the output waveform of half/full wave rectifier and calculate its ripple factor and efficiency. (C141.4)
6. To study the reverse bias I-V (current-voltage) characteristics of a Zener diode. Also determine the breakdown voltage, static and dynamic resistances. (C141.4)
7. To plot input characteristics of a common emitter NPN BJT. (C141.2)
8. To plot output characteristics of a common emitter NPN BJT. (C141.2)
9. To realize inverting and non amplifier configuration using Op-Amp IC- 741. (C141.4)
10. To realize adder and subtractor circuits using Op-Amp IC-741. (C141.4)
11. Verification of Superposition Theorem. (C141.3)
12. Realization of desired wave shapes using clipper and clamper circuits. (C141.4)

Virtual Lab:

1. To plot input characteristics of a common collector NPN BJT. (C141.2)
2. To plot output characteristics of a common collector NPN BJT. (C141.2)

PREFACE

The Basic Electronics Lab is one of the most important labs for electronics engineering students. This is the first level laboratory in which students are introduced with electronics for the first time and are trained with the basics of circuit analysis in electronics. In this lab students are given introduction and identification of different active and passive electronics devices/components. They are familiarized with different signal sources and instruments used in electronics engineering, students learn the usage/handling of different measuring instruments like Digital Multimeter, DSO, Arbitrary Function Generators, Regulated power supply etc. In this lab, students become familiar with device characteristics and their applications. Students develop circuits on bread board and analyze it using laboratory instruments such as digital multimeters (DMMs), power supplies, function generators and oscilloscopes. By the end of their lab coursework, students become familiar with basic components such as resistors, diodes, transistors, and operational amplifiers function and have the necessary skills of circuit designs.

A practical approach is probably the best approach to mastering a subject and gaining a clear insight. List of experiments of Basic Electronics Lab covers those practical oriented electronic circuit that are very essential for the students to solidify their theoretical concepts. This provides a communication bridge between the theory and practical of the electronic circuits. This is a one of most challenging laboratories in terms of utilization because students of all departments use this lab.

Lab is equipped with sufficient number of following Equipments

- Digital Multimeter,
- Regulated Power Supply,
- DSO,
- Arbitrary Signal Generator,
- Project Bread Board

GUIDELINES AND PRECAUTIONS FOR STUDENTS

1. General Guidelines:

- Be punctual while coming to the lab and be sincere in doing your lab work.
- Before starting the lab experiment, you must go through the lab manual of the respective experiment and get your doubts clear before starting the experiment.
- Be careful to keep watch bands, rings, necklaces and the other metallic object out of contact with live parts when working around electrical apparatus.
- If you are not sure how to operate a piece of equipment then read the manual or take help from lab staff/ teacher.
- No food, beverages, chewing tobacco and gums are allowed in the lab.
- At the end of each lab period, return all the components and instruments that are issued to you or your group.
- Keep the work area neat and clean; arrange the stools before leaving the laboratory.

2. Precautions to be taken while doing experiment:

- Understand the circuit and equipment to be used in the experiment.
- Make the circuit on the bread board and check for all the connections before switching 'ON' the supply.
- Use the proper source (A.C or D.C) and the range of meters. Always try to vary the voltage gradually.
- Never exceed the permissible values of current, voltage of any machine, apparatus, wires, load etc.
- Do not leave loose wire (wires not connected).
- Switch off the circuit while making any modification in the circuit.
- When working with inductive and capacitive circuits, reduce voltages or current to near zero before switching open the circuits.
- While connecting the CRO and Function Generator with the circuits, always connect with proper polarity of the BNC cable.

EXPERIMENT NO. -1

AIM:

Introduction to various components (Resistor, Capacitor, inductor, and IC) and instruments Multimeter, Bread board and Regulated D.C. power supply, Function Generator and CRO.

THEORY:

Different Resistor Types:

Carbon film resistors:

The size of the resistor decides its power rating (i.e., the maximum power it can dissipate without burning).

Power rating from the top of the graph: 1/8 W

1/4 W

1/2 W



Metal film resistors: Used when a higher tolerance (more accurate value) is needed.



Power rating from the top of the graph:

1/8 W (tolerance $\pm 1\%$) 1/4 W (tolerance $\pm 1\%$) 1 W (tolerance $\pm 5\%$) 2 W (tolerance $\pm 5\%$)

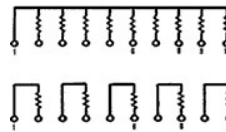
Reading resistor values from the colored bands:

0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9									
0	Black								
1	Brown								
2	Red								
3	Orange								
4	Yellow								
5	Green								
6	Blue								
7	Purple								
8	Grey								
9	White								
±5%		Gold							
±10%		Silver							
Color Codes									

Brown ±1% Red ±2% Gold ±5% Silver ±10%*									
0 X1 1 1 X10 2 2 X100 3 3 X1000 4 4 X10000 5 5 X100000 6 6 X1000000 7 7 ÷10 8 8 ÷100 9 9									
27K EXAMPLE									
4 Band Resistors									

Brown ±1% Red ±2% Gold ±5%* Silver ±10%*									
0 0 X1 1 1 1 X10 2 2 2 X100 3 3 3 X1000 4 4 4 X10000 5 5 5 ÷10 6 6 6 ÷100 7 7 7 8 8 8 9 9 9									
15K EXAMPLE									
5 Band Resistors									

Single-In-Line (SIL) Resistor network:



Variable Resistors:



Wirewound resistors:



<http://www.hobby-electronics.org/>

Ceramic (or cement) resistor:



Thermistor (thermally sensitive resistor):



SMD resistors (Surface-Mount Device)



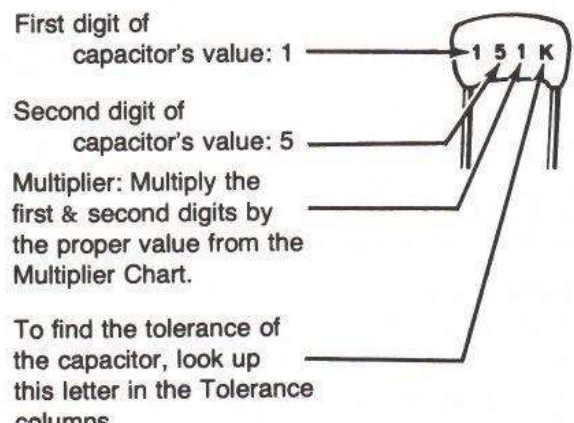
Different Capacitor Types:

Ceramic Capacitors: Limited to quite small values, but have high voltage ratings. They range from 1pF to 0.47 μ F and are not polarized.



Reading Ceramic Capacitor values:

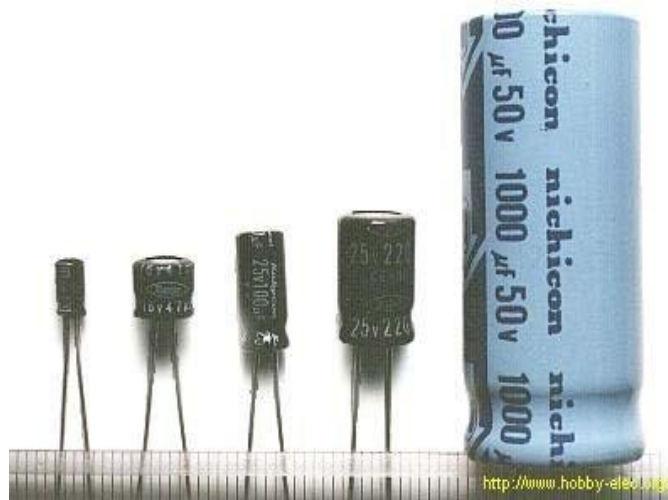
Example: **102** means 10 (and two zeroes) 00 or 1,000 pF or .001 μ F.



For the number:	Multiply by:	LETTER	TOLERANCE 10pF or LESS	TOLERANCE OVER 10pF
0	1	B	+/- 0.1pF	
1	10	C	+/- 0.25pF	
2	100	D	+/- 0.5pF	
3	1000	F	+/- 1.0pF	+/- 1%
4	10,000	G	+/- 2.0pF	+/- 2%
5	100,000	H		+/- 3%
		J		+/- 5%
8	0.01	K		+/- 10%
9	0.1	M		+/- 20%

Electrolytic Capacitors (Electrochemical type capacitors): Used for all values above 0.1 μ F. Electrolytics have lower accuracy and temperature stability than most other types and are almost always polarised. It's usually best to only use an electrolytic when no other type can be used, or for all values over 100 μ F.

From the left to right: 1 μ F (50V)
 47 μ F (16V)
 100 μ F (25V)
 220 μ F (25V)
 1000 μ F (50V)



Tantalum Capacitors: Tantalum capacitors pack a large capacity into a relatively small and tough package compared to electrolytics, but have much smaller voltage ratings. They are often polarized and range from 0.1 μ F to 100 μ F.

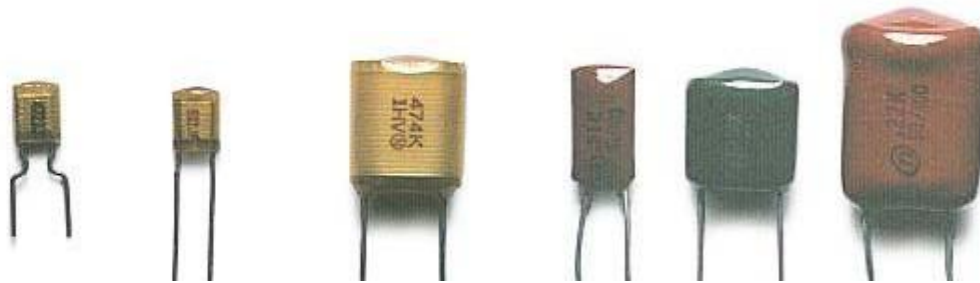
From the left to right: 0.33 μ F (35V)



0.47 μF (35V)

10 μF (35V)

Polyester Film Capacitors (Green Caps): Ranging from 0.01 μF to 5 μF . They are similar to ceramics with some larger values and a slightly larger construction. They are not polarized.



Metallized Polyester Film Capacitors:



SMD Capacitors:



Variable Capacitors:



Different Inductor Types:

Inductors:



Reading Inductor values from color codes:

Result Is In μH

4-BAND-CODE $\Rightarrow$  $\Rightarrow$ $270\mu\text{H} \pm 5\%$

COLOR	1st BAND	2nd BAND	MULTIPLIER	TOLERANCE
BLACK	0	0	1	$\pm 20\%$
BROWN	1	1	10	Military $\pm 1\%$
RED	2	2	100	Military $\pm 2\%$
ORANGE	3	3	1,000	Military $\pm 3\%$
YELLOW	4	4	10,000	Military $\pm 4\%$
GREEN	5	5		
BLUE	6	6		
VIOLET	7	7		
GREY	8	8		
WHITE	9	9		
NONE				Military $\pm 20\%$
GOLD			0.1 / Mil. Dec. Pt.	Both $\pm 5\%$
SILVER			0.01	Both $\pm 10\%$

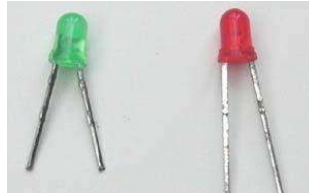
Military Identifier $\Rightarrow$  $\Rightarrow$ $6.8\mu\text{H} \pm 10\%$
MILITARY CODE

Other interesting components:

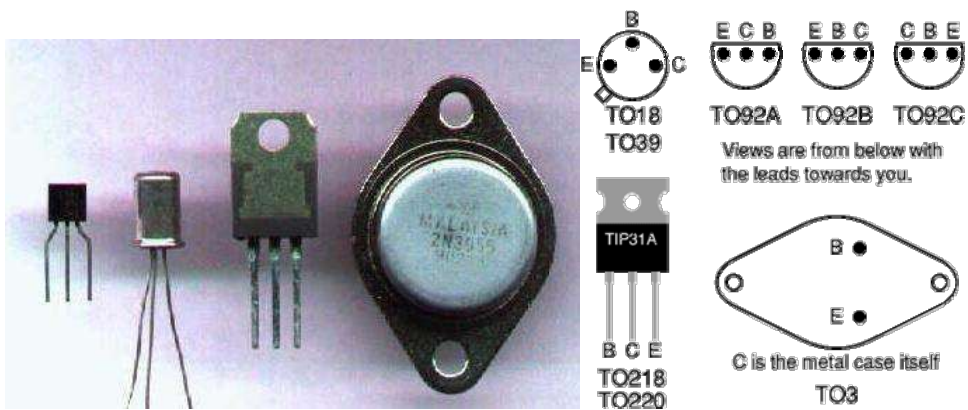
Diodes:



LED (Light Emitting Diodes):



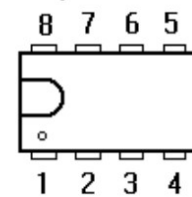
Transistors:



ICs (Integrated Circuits):



component side



Introduction to CRO

The cathode-ray oscilloscope (CRO) is a common laboratory instrument that provides accurate time and amplitude measurements of voltage signals over a wide range of frequencies. Its reliability, stability, and ease of operation make it suitable as a general purpose laboratory instrument. The heart of the CRO is a cathode-ray tube shown schematically in Fig. 1.

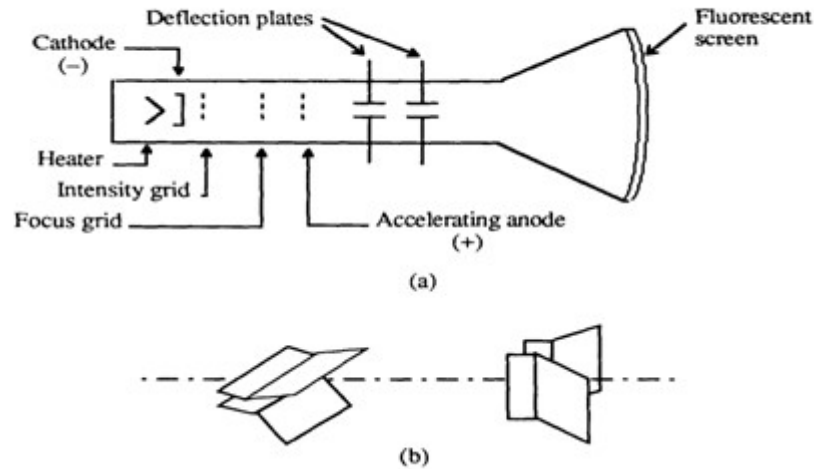


Figure 1: Cathode Ray Tube (a) Schematic, (b) Details of Deflection Plate

Electrons leave the **heated cathode** by thermionic emission. They are accelerated through a fixed voltage and emerge as a narrow beam focused through a hole in the **deflection plate**. When the **electron beam** strikes the **fluorescent screen** on the face of the tube, it produces a small luminous spot. An external potential difference can be measured by applying it across a pair of parallel **deflecting plates**, through which the beam passes on the way to the screen. The beam is then deflected by the resultant transverse uniform electric field between the plates. There are two pairs of deflecting plates, one for vertical and the other for horizontal deflection.

CRO Controls

The front panel of oscilloscope is shown in figure 2. The inputs and controls with which you will be concerned can be divided into three groups depending on the part of the oscilloscope circuitry with which they are associated. All oscilloscopes include this basic set of controls. The numbers in the diagram refer to the controls described below.

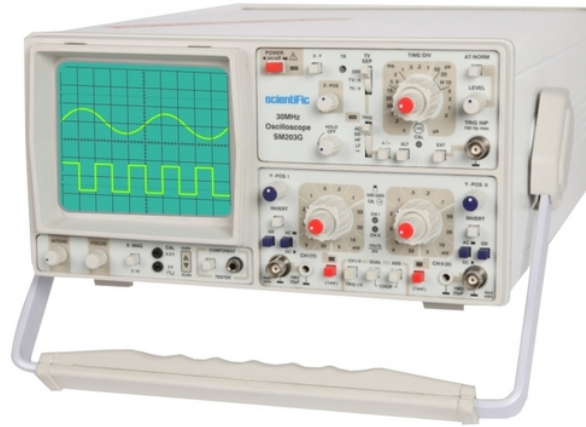


Figure 2: CRO Controls

(I) FORMATION OF THE ELECTRON BEAM

POWER : Push in to turn the power on.

BEAM INTENSITY : The INTEN control adjusts the intensity or brightness of the trace. Clockwise rotation increases the intensity.

IMPORTANT NOTE: When the spot on the screen is stationary, keep the intensity very low in order not to damage the fluorescent screen at that point. In general, do not leave the intensity higher than necessary for reasonable visibility.

BEAM FOCUS : The FOCUS control adjusts the sharpness of the electron spot on the screen.

(II) VERTICAL DEFLECTION OF THE ELECTRON BEAM

VERTICAL AXIS SENSITIVITY: VARIABLE FINE ADJUSTMENT: The vertical axis VARIABLE control is used for fine adjustments of the vertical axis sensitivity. The sensitivity is reduced by turning the vertical axis VARIABLE knob counterclockwise. Only in the fully clockwise position (with the knob clicked into this position) can the vertical sensitivity be read from the VOLTS/DIV setting.

VERTICAL POSITION: This control is used to move the displayed trace up or down. Turning the knob clockwise moves the trace upward.

AC-DC INPUT SELECTOR: With the button in (DC position), the input terminal is directly coupled to the vertical amplifier. With the button out (AC position) the direct current is blocked by a capacitor.

GND GROUND SWITCH: With the GND button in, the input to the vertical amplifier is grounded and the vertical INPUT terminal has zero input voltage.

VERTICAL INPUT : This is the input terminal for the vertical amplifier. The maximum permissible input voltage is 120 Volts.

VERTICAL AXIS SENSITIVITY RANGE SELECTION IN VOLTS/DIV: The VOLTS/DIV 11-position switch determines the sensitivity of the vertical amplifier. The 11 ranges are indicated in volts per division on the front panel. The indicated sensitivities are only correct if the vertical axis VARIABLE control is in the calibrated position (fully clockwise) and the VERTICAL position control is pushed in.

(III) HORIZONTAL DEFLECTION OF THE ELECTRON BEAM

EXT TRIG INPUT or H IN: EXTERNAL TRIGGER INPUT OR HORIZONTAL INPUT: This input is used for externally triggering the horizontal sweep or for the HORIZONTAL INPUT in a x-y plot. This input is DC coupled (i.e., either a constant or a time varying voltage may be used -- compare with above) and the maximum allowable applied voltage is 100 volts.

HORIZONTAL SWEEP TRIGGER CONTROLS: There are four switches and one knob that are used in determining the conditions for triggering the horizontal sweep.

HORIZONTAL SWEEP TIME: VARIABLE FINE ADJUSTMENT: The horizontal axis VARIABLE control is used for fine adjustments of the horizontal sweep time per division. The sweep time per division is increased by turning the horizontal axis VARIABLE knob counter clockwise. Only in the fully clockwise position (with the knob clicked into position) can the sweep time be read from the TIME/DIV setting.

HORIZONTAL SWEEP TIME: RANGE SELECTION IN TIME/DIV or H IN HORIZONTAL INPUT SELECTOR: This control is used to select the horizontal sweep time per division on the scope face. The indicated sweep times per division are only correct if the horizontal axis VARIABLE control is in the calibrated position (fully clockwise) and the HORIZONTAL position control is pushed in (no magnification position). In the fully counterclockwise position the sweep oscillator is disconnected and the signal on the H IN terminal is applied to the horizontal plates.

HORIZONTAL POSITION: This control is used to move the trace left or right. Turning the knob clockwise moves the trace to the right.

Introduction to Function Generator

In the laboratory the device we use to produce A.C. signals of various types is called the Waveform or Function Generator. Figure 3 shows the function generator controls. Display on its own tells us nothing of the shape or amplitude of the output. It only gives a number which must

be multiplied by the range settings to find the signal frequency. To set the frequency output of the Function Generator we use the Frequency Control and the Range Selection Switches.



Figure 3 : Function Generator

1. The process is quite simple, firstly the Frequency is adjusted to the correct value. i.e if we require 12.5 kHz we Use the Frequency Control to 12.5 and then depress the Range Selection Switch corresponding to a 1kHz multiplier and the output will be as required.
2. The Mode Selection Switches are used to control the shape of the output signal. In this case we only have access to three different types of signal: Square Wave, Sine Wave and Triangular wave.
3. The operation of these switches is quite simple just press the switch that corresponds to the particular type of signal that we require.
4. Before we connect the output of the Function Generator to a circuit we must first be able to set the Amplitude of the Output signal.
5. The Output Terminal on the Function Generator is a BNC Connection which can be used directly with a Coaxial Cable. Or if 3mm Plugs are required we will need to use an adaptor.

Introduction to DC Power Supply

It is easy to use, low power, four output general purpose laboratory supply. It is suitable for experimental set-ups, circuit development and low voltage applications. The power supply delivers three dc outputs through section A, B and C. All the sections are provided with panel meters to monitor output voltage and current separately.



Figure 4: DC Regulated Power Supply

- Section A : 0-30 V dc output with 2 Amps capacity.
- Section C : 5V preset dc output with 5 Amps capacity.
- Section B : 0 to +/-15V preset dc output with 1A capacity.

All the outputs are floating (i.e. neither any of +ve or -ve output terminals nor any point within the regulator circuitry is connected to ground). In the other words all the three sections can be used simultaneously. The power supply is designed to operate satisfactorily in ambient temperature of up to 50°C and full power can be drawn if free air circulation is allowed. The unit works on main supply of 230v ac at 50 Hz.

Introduction to Digital Multimeter (DMM)

The DMM (**Figure 5**) is used to make measurements of Voltage, Current and Resistance. Both A.C. and D.C. signals can be measured using the DMM.

1. There is a dial that is used to select the mode which we wish to make measurements in.
2. Below are the Modes listed with their corresponding numbers from the diagram:
 - a. D.C Voltage measurement.
 - b. A.C Voltage measurement.
 - c. D.C Current measurement.
 - d. A.C Current measurement.
 - e. Resistance measurement.



Figure 5: Digital Multimeter

3. Terminals are labelled Terminal1 through to Terminal4.
 - a. 10A: This terminal is used for measuring signals in the range of 2-10 Amps.
 - b. mA: This is used to measure signals from 0 to 400 mAmps.
 - c. VΩHz ~: This is used when we measure voltage/ resistance etc.
 - d. COM: This is the common terminal and is used in all measurements.
4. DMM is connected in parallel for voltage and in series for current.
5. DMM gives different readings than a CRO for A.C. signals.
6. Always connect the COM terminal when making measurements.

Introduction to Breadboard

A breadboard (Figure 6) is used to make up **temporary circuits** for testing or to try out an idea. No soldering is required so it is easy to change connections and replace components. Parts will not be damaged so they will be available to re-use afterwards

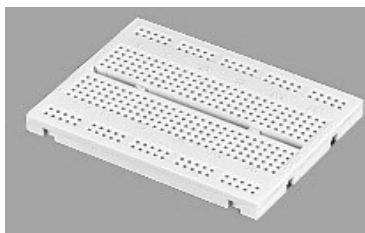


Figure 6: Project Bread Board

Breadboards have many tiny sockets (called 'holes') arranged on a 0.1" grid. The leads of most components can be pushed straight into the holes. ICs are inserted across the central gap with their notch or dot to the left. The top and bottom rows are linked **horizontally** all the way across as shown in figure 7. The power supply is connected to these rows, + at the top and 0V (zero volts) at the bottom. The other holes are linked **vertically** in blocks of 5 with no link across the centre as shown in figure 7.

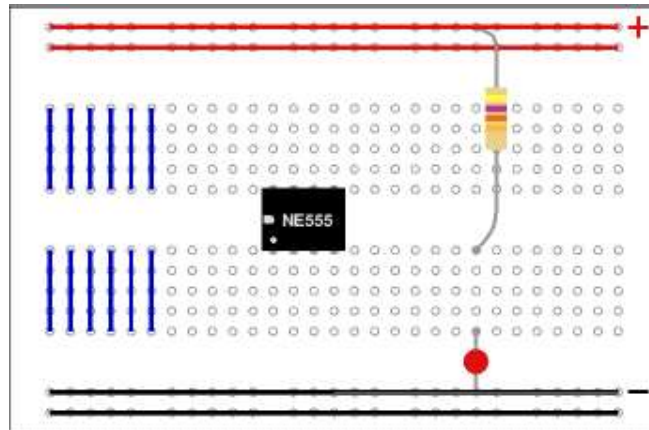


Figure 9

Brief Introduction about DSO

A digital storage oscilloscope (DSO) is an electronic instruments that measures and records electrical signal. It converts the analog signal into a digital format and stores it in its digital memory, allowing for easy recall and analysis.

Digital storage oscilloscopes are used in a variety of applications. Biomedical engineers use them to measure electrical signals from the human body, such as heartbeats and brainwaves. Telecommunications engineers use DSOs to test cell phones and other communication devices. Aerospace engineers use them to test aircraft components, while automotive engineers use them to test vehicle electronic systems. Design engineers use DSOs to verify the performance of their designs. [KeySight]



Figure 10

A digital storage oscilloscope typically has five main components:

1. Display
2. Vertical input channel
3. Horizontal input channel
4. Trigger
5. Analog-to-digital converter

The display is used to visualize the electrical signal being measured. The vertical input channel is used to measure the amplitude of the signal, while the horizontal input channel is used to measure the signal's frequency. The trigger is used to start and stop the measurement process.

Digital storage oscilloscopes have two operating modes: single-shot and repetitive.

In single-shot mode, the oscilloscope acquires and stores one signal. In repetitive mode, the oscilloscope continuously acquires and stores signals.

Brief Introduction about Advanced Arbitrary Waveform / Function Generator

Function generator offers the standard signals and features you expect — modulation, sweep, and burst. It also has features that give you the capabilities and flexibility you need to get your job done quickly, no matter how complex.

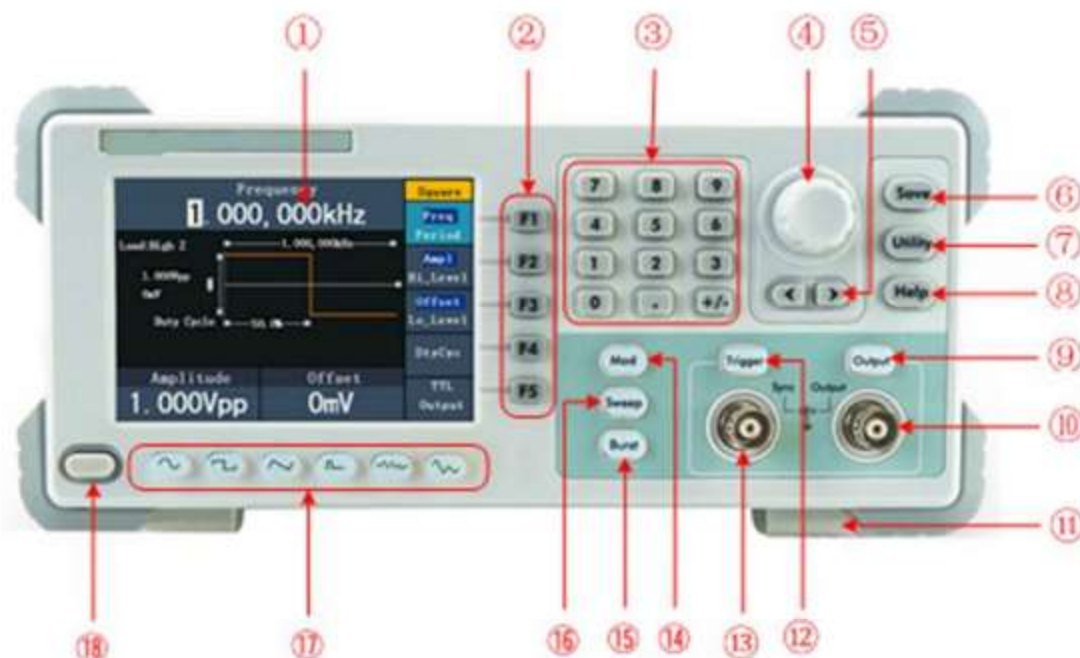


Figure 11

The Trueform function generator offers the highest signal fidelity so you can generate the exact waveforms you need for your most challenging measurements. You can be confident you are seeing your design's characteristics, and not that of your digital waveform generator, in your

measurements. [Keysight]

①	LCD	Display the user interface
②	Menu selection buttons	Include 5 buttons: F1 - F5, activate the corresponding menu
③	Number keys	Input parameters, include: number, point and plus/minus sign
④	Knob	Change the current highlighted number, also can be used to select file location or switch the character of the soft keyboard when entering file name. In Sweep and N-Cycle Burst, if you choose Source as "Manual", every time you press this knob, the generator will be triggered once.
⑤	Direction key	Move the cursor of the focused parameter or select the file locations
⑥	Save button	Manage the waveform files in the file system
⑦	Utility button	Set the auxiliary system function.
⑧	Help button	View the build-in help information
⑨	Output key	Activate or deactivate the output signal. The button is lit when it is in the On state.
⑩	Main Output Terminal	Output main signal

EXPERIMENT NO: 2

AIM:

Verification of KVL and KCL using a given circuit.

APPARATUS AND COMPONENTS REQUIRED:

Sl.No	Apparatus	Specifications	Quantity
1	Multimeter	Digital	1
2	Power Supply	DC regulated	1
3	Bread Board	--	1
5	Connecting wire	-	
5	Resistors	1 K Ω	2
6	Resistors	4.7 K Ω	1

THEORY:

There are two Kirchhoff Laws: (a) Kirchhoff Voltage Law (KVL); and (b) Kirchhoff Current Law (KCL).

KVL states that the algebraic sum of all the voltages encountered as one goes around a complete loop is zero. The word *algebraic* implies that the polarity of each of the voltages is duly taken into account. The application of KVL is illustrated using the simple circuit given in Figure 1 where the existence of three loops is readily identified so that one can write down the three equations given in Eq. (E2.1).

CIRCUIT DIAGRAM:

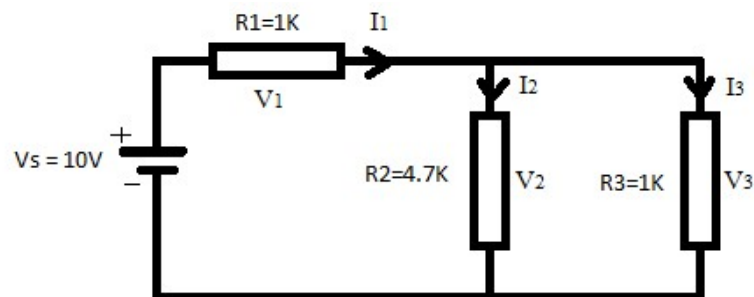


Figure 1: A simple circuit for illustrating KVL and KCL

$$V_s - R_1 I_1 - R_2 I_2 = 0 \quad (\text{E2.1a})$$

$$R_2 I_2 - R_3 I_3 = 0 \quad (\text{E2.1b})$$

$$V_s - R_1 I_1 - R_3 I_3 = 0 \quad (\text{E2.1c})$$

Although KVL yields three simultaneous equations in three unknowns, only two of the three equations are independent. This you can prove by doing a lit bit of algebra and deriving one of the equations from the other two. When solving simultaneous equations in three unknowns, one needs three independent equations in the three unknowns. Two are provided by KVL and the third by KCL (Kirchhoff Current Law) enunciated in Sect. 2.2. below.

Kirchhoff Current Law (KCL)

KCL states that the algebraic sum of all the currents entering (or leaving) a junction or node is zero. This law is simply a statement of the physical requirement that charge cannot simply accumulate at a node point; it simply must keep moving. This is entirely analogous to the hydraulic problem in which a pipe carrying water branches out, say, into two pipes; clearly the total incoming water flow in the first pipe must equal the sum of the outgoing water flows in the other two pipes, i.e., water simply does not accumulate at the junction point. The equation yielded by KCL applied to the top junction of three branches in Figure: 1 is

$$I_1 - I_2 - I_3 = 0 \quad (E2.2)$$

Note that since currents are algebraic quantities, one can assign either direction to each of the currents.

PROCEDURE:

1. Measure the practical value of the given resistance with the help of Digital multimeter.
2. Make the circuit on the breadboard as per the given circuit diagram,
3. Measure the value of input voltage and the voltage across each resistor (R_1 , R_2 , R_3) with the help of DMM.
4. Measure the current I_1 , I_2 and I_3 with the help of DMM.

NOTE:- To measure current, Ammeter is always connected in series with the circuit. So break the circuit at that point and create two points. Now insert the Ammeter between given two points to measure current.

OBSERVATION TABLE:

Table I – Resistance Measurements

Value Determination	R_1	R_2	R_3
ColorCode			
Measured			

Table II – Voltage Supply Measurements

Value Determination	V_s
Nominal	10 V
Measured	

Table III – MEASURED and THEORETICAL Currents and Potentials

For the different values of input voltage V_s , measure the currents through the resistors and voltage drops across the resistors.

	V_s (Input Voltage) (Volts)	V₁ (Voltage across R₁) (Volts)	V₂ (Voltage across R₂) (Volts)	V₃ (Voltage across R₃) (Volts)	I₁ (Current through R₁) (mA)	I₂ (Current through R₂) (mA)	I₃ (Current through. R₃) (mA)
Measured							
Theoretical							

CALCULATION:

Put the values of voltages and currents in equations (E2.1a – E2.1c and E2.2)

Verify that in each loop

$$\Sigma V_i = 0$$

And at the node

$$\Sigma I_i = 0$$

Also calculate for the theoretical values.

RESULT:

PRECAUTION:

LEARNING OUTCOME:

EXPERIMENT NO : 3

AIM:

Verification of Thevenin's Theorem.

APPARATUS AND COMPONENTS REQUIRED:

Sl. No	Apparatus	Specifications	Quantity
1	Multimeter	Digital	1
2	DC Regulated Power Supply	10V	1
3	Bread Board		1
4	Connecting wire		
5	Resistors	1 K Ω , 2.2 K Ω , 3.3 K Ω , 4.7 K Ω , 10 K Ω	1 each

THEORY:

Thevenin equivalents are circuit simplification techniques that focus on terminal behavior. They are especially useful in analyzing power systems and other circuits where one particular resistor in the circuit (called the load resistor) is subject to change, and recalculation of the circuit is necessary with each trial value of load resistance, to determine voltage across it and current through it. They are extremely valuable aids in analysis. Their equivalent circuits may be used to represent any circuit made up of linear elements. These two theorems are two equally valid methods of reducing a complex network down to something simpler to analyze. A Thevenin equivalent circuit can be readily converted to a Norton equivalent circuit and, vice versa. Figure 1 pictures Thevenin equivalents of a circuit.

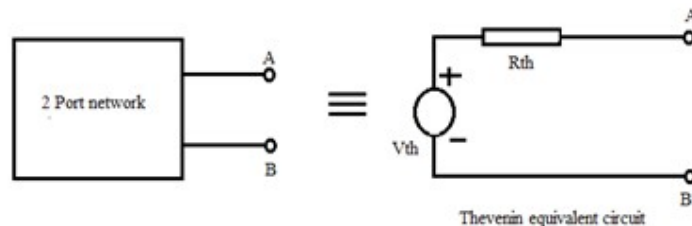


Figure 1 : A Thevenin equivalent circuit

Thevenin's Theorem states that we can replace entire network, exclusive of the load, by an equivalent circuit that contains only an independent voltage source in series with an impedance (resistance) such that the current-voltage relationship at the load is unchanged .

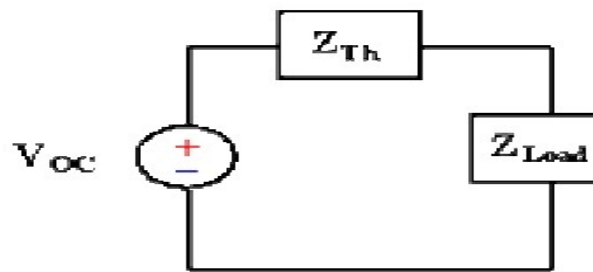


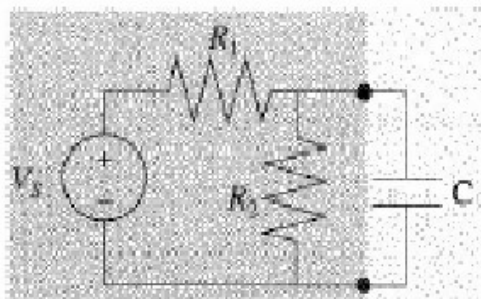
Figure 2: Equivalent Circuit of Thevenin's Theorem

For a given complex electrical circuit, following steps are done in order to find Thevenin and Norton equivalents of that circuit.

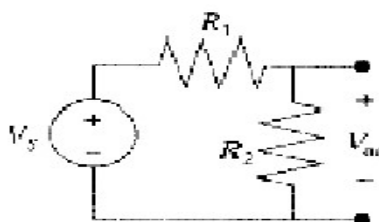
1. Pick a good breaking point in the circuit (cannot split a dependent source and its control variable).
2. Compute the open circuit voltage, V_{OC} .
3. Compute the Thevenin equivalent resistance, R_{Th} (or impedance, Z_{Th}).
 - a. If there are only independent sources, then short circuit all the voltage sources and open circuit the current sources (just like [superposition](#)).
 - b. If there are only dependent sources, then must use a test voltage or current source in order to calculate $R_{Th} = v_{Test}/i_{Test}$ (or $Z_{Th} = V_{Test}/I_{Test}$).
 - c. If there are both independent and dependent sources, then compute R_{Th} (or Z_{Th}) from $R_{Th} = v_{OC}/i_{SC}$ (or $Z_{Th} = V_{OC}/I_{SC}$).
4. Replace circuit with Thevenin equivalent.
 V_{OC} in series with R_{Th} (or Z_{Th})

Note: for figure 2 the equivalent network is merely R_{Th} (or Z_{Th}), that is, no current or voltage sources.

Let's take a look in the following circuit and try to find Thevenin equivalents for the shaded part of it.



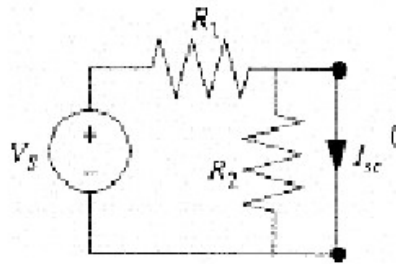
First, we remove the capacitor from the circuit; we can re-introduce it once we have derived the equivalent circuit.



The open-circuit voltage V_{oc} is found from the voltage divider rule:

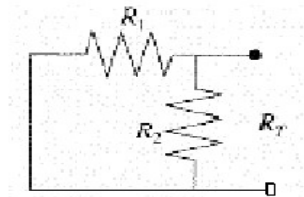
$$V_{oc} = V_s \frac{R_2}{R_1 + R_2}$$

The short-circuit current I_{sc} is found by determining the current flowing between the terminals when they are shorted:



$$I_{sc} = \frac{V_s}{R_1}$$

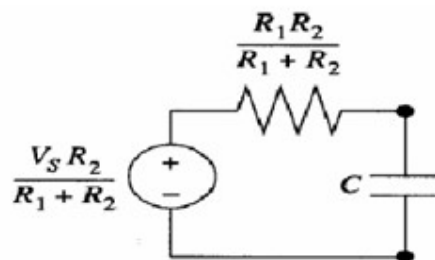
The equivalent resistance can be found by de-activating the independent voltage source by replacing it by a short circuit, as shown:



$$R_T = \frac{R_1 R_2}{R_1 + R_2}$$

We can verify the standard relationship $V_{oc} = I_{sc} \times R_T$ is satisfied: hence, we could have derived only two of these three parameters.

Consequently, the Thevenin equivalent circuits are as follows:



CIRCUIT DIAGRAM:

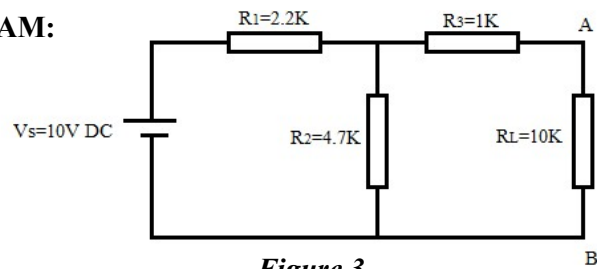


Figure 3

PROCEDURE:

1. Measure the practical value of given resistance.
2. Make the circuit on the breadboard as per the circuit diagram in Figure 3
3. Measure the voltage (V_L) across the load resistor (R_L) and the current I_L through the Load using DMM Figure 4.

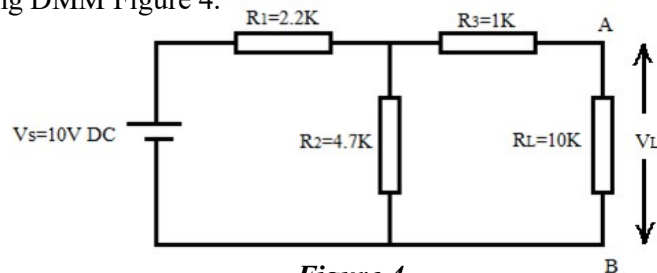


Figure 4

4. The DC supply is switched OFF and the Load R_L is removed Figure 5.

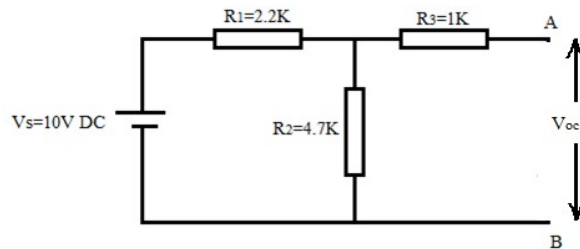


Figure 5

5. The DC supply is switched ON and the voltmeter is connected between the port A and B to measure the Thevenin voltage ($V_{th} = V_{oc}$) as shown in the Figure 5.
6. The power supply is switched off, voltage source is replaced by a connecting wire (short circuit or internal resistance of the source). Using Ohm-meter the resistance is found between A and B which is Thevenin resistance R_{th} Figure 6.

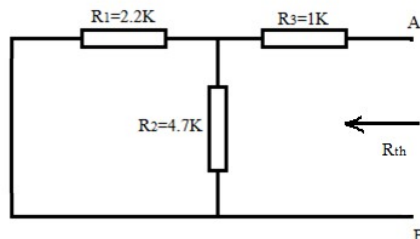


Figure 6

7. The circuit is reconnected and replaced by a single voltage source V_{th} and an equivalent resistance R_{th} in series with the Load R_L . Figure 7.

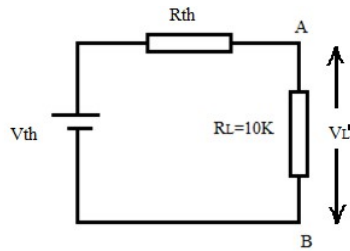


Figure 7

8. Measure voltage across $R_L = V_L'$ and current I_L' through R_L
 $V_L = V_L'$
 $I_L = I_L'$

OBSERVATION TABLE:

Table I – Resistance Measurements

Value Determination	R_1	R_2	R_3
Color Code			
Measured			

Table II – Voltage Supply Measurements

Value Determination	V_s
Nominal	10 V
Measured	

Table III – MEASURED Currents and Potentials

	V_L	$V_{oc} = V_{th}$	R_{th}	V_L'
Measured				
Theoretical				

CALCULATION:

Student will do calculation as outlined above in theory and compare with experimental value

RESULT:

PRECAUTION:

LEARNING OUTCOME:

EXPERIMENT NO. : 4

AIM:

To study the forward and reverse bias (volt-ampere) characteristics of a simple p-n junction diode. Also determine the forward resistance of the diode.

APPARATUS AND COMPONENTS REQUIRED:

Sl. No	Apparatus	Specifications	Quantity
1	DC Power Supply	-	1
2	Bread Board	-	1
3	Multimeter	-	1
4	Connecting wire	-	-
5	PN Diode	IN4007	1
6	Resistors	1K Ω	1

THEORY:

The diode is a semiconductor device formed from a junction of n-type and p-type semiconductor materials. The lead connected to p-type material is called as anode and the lead connected to n- type material is called cathode as shown in Figure 1.



Figure 1: (a) Diode Symbol (b) Actual diode

When diode is in forward bias (higher potential is connected to the anode lead), current flows through it. As the voltage across diode increases, current also increases. It is observed that the rate of change in current increases with increase in the voltage across diode. After some potential drop across the diode, the rate of change in current increases rapidly. The potential drop after which current increases rapidly is called as **Cut in Voltage**. It is measured by drawing a tangent on the slope of the V-I Characteristics from where current increases drastically. For Germanium diode it is around 0.2-0.3V, while for Silicon diode it is 0.4-0.7V.

When diode is in reverse bias (higher potential is connected to the cathode lead), ideally no current flows through it, practically a very small leakage current (in micro ampere) flows due to some impurity charge carriers. In the p-n junction diode the value of reverse breakdown voltage is very high (-600V for 1N4001). If this voltage is applied in the reverse bias, then very high value of current will flow and diode will get damaged. The characteristic curve for an ideal diode and practical diode is shown in Figure 2.

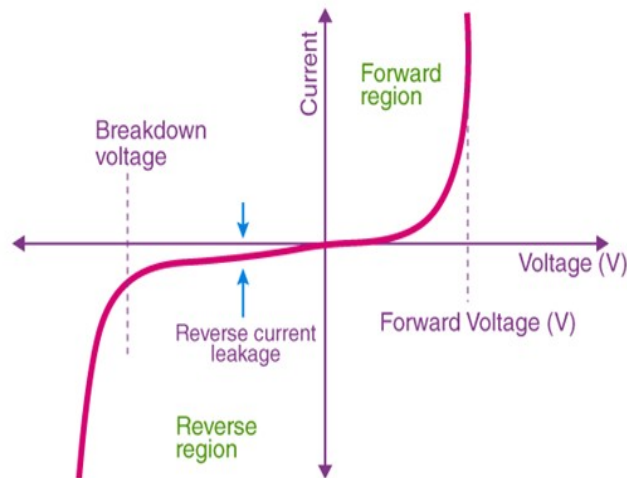


Figure 2: V-I characteristics of Diode

While analyzing circuits, the practical diode is usually replaced with a simpler model. In the simplest form, the diode is modeled by a switch as shown in Figure 3. The switch is closed when the diode is forward biased and open when reverse biased. R_f and R_r are the resistances offered by the diode in the forward bias and reverse bias respectively. V_y is the cut-in voltage of the diode.

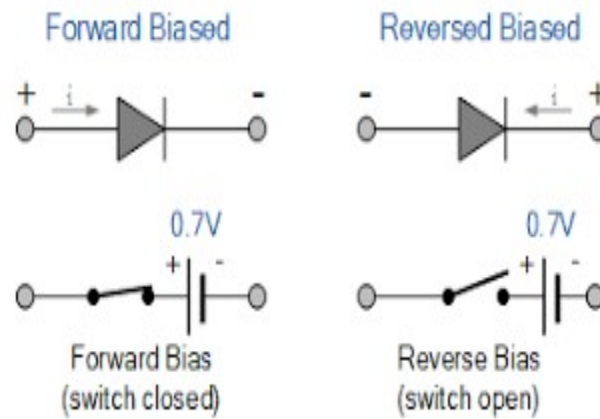


Figure 3: Equivalent model of diode

CIRCUIT DIAGRAM:

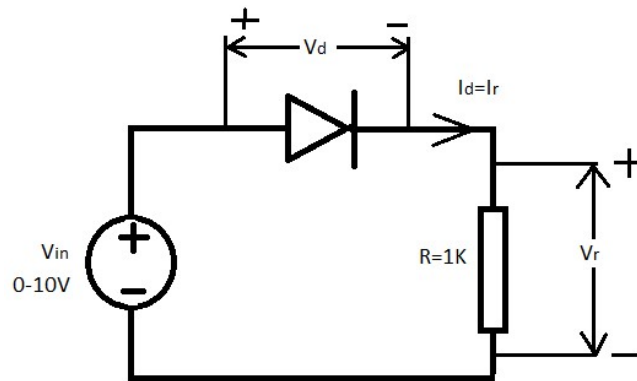


Figure 4: Circuit diagram

PROCEDURE:

1. Study the characteristics of the diode in the data sheet. Copy all the specification for the diode in your final lab report.
2. Connect the circuit as shown in fig. 4.
3. Connect voltmeter across diode and resistor respectively.
4. Vary the input from DC source from -5 to 5 volt in the step of 0.1 volt and note down the reading across diode and resistor. Calculate the current flowing through the resistance.
5. Now plot the graph between the current flowing through the diode and potential across the diode.
6. From this graph, obtain the cut-in voltage (V_y) for the diode. From the point where current increases sharply, draw a line to x-axis. The point where it intersects on the x-axis is called cut-in voltage.
7. Measure the slope of the curve to calculate the resistance offered by the diode in the forward bias. Forward bias resistance is named as R_f as shown in fig. 6.

OBSERVATION TABLE:

S. No.	Input Voltage (V_{in})	Resistor Voltage (V_r)	Current ($I=V_r/R$)	Diode Voltage (V_d)

RESULT:

$V_y =$

$R_f =$

LEARNING OUTCOME:

EXPERIMENT NO.: 5

AIM:

To observe the output waveforms of half wave and full wave rectifiers and calculate its ripple factor and efficiency.

APPARATUS AND COMPONENTS REQUIRED:

Sl. No	Apparatus	Specifications	Quantity
1	CRO	-	1
2	Bread Board	-	1
4	Connecting wire	-	-
5	Transformer	(12-0-12), 500mA	1
6	Diode	IN4007	4
7	Resistors	1K Ω	1

THEORY:

Half wave and full wave bridge rectifier:

In half wave rectification, single diode act as a half wave rectifier. The A.C. supply to be rectified is applied in series with diode and load resistance R_L . A.C. supply is given through a transformer. During positive half cycles of input A.C. voltage, this make diode forward biased and hence it conducts current. During negative half cycles, diode is reverse biased and it conducts no current. Therefore, current flow through the diode during positive half cycles of A.C. input voltage only. it is blocked during negative half cycles.

Full wave bridge rectifier, current flow through the load in the same direction for both half cycles of input A.C. it's contain four diode D1, D2, D3, D4 connected to form bridge. The A.C supply to be rectified is applied to the diagonally opposite end of the bridge trough the transformer. Between two other ends of the bridge, the load resistance R_L is connected. During the positive half cycle of A.C. input voltage D1 & D3 forward biased while diode D2 & D4 are reversed biased. Therefore only diode D1& D3 conduct. These two diode will be in series through the load. Current flow through load. During negative half cycles diodes D2&D4 becomes forward biased whereas D1 and D3 become reverse biased. Therefore only D2&D3 conduct .these diodes will be in series through the load R_L . Current flow through lo

CIRCUIT DIAGRAM:

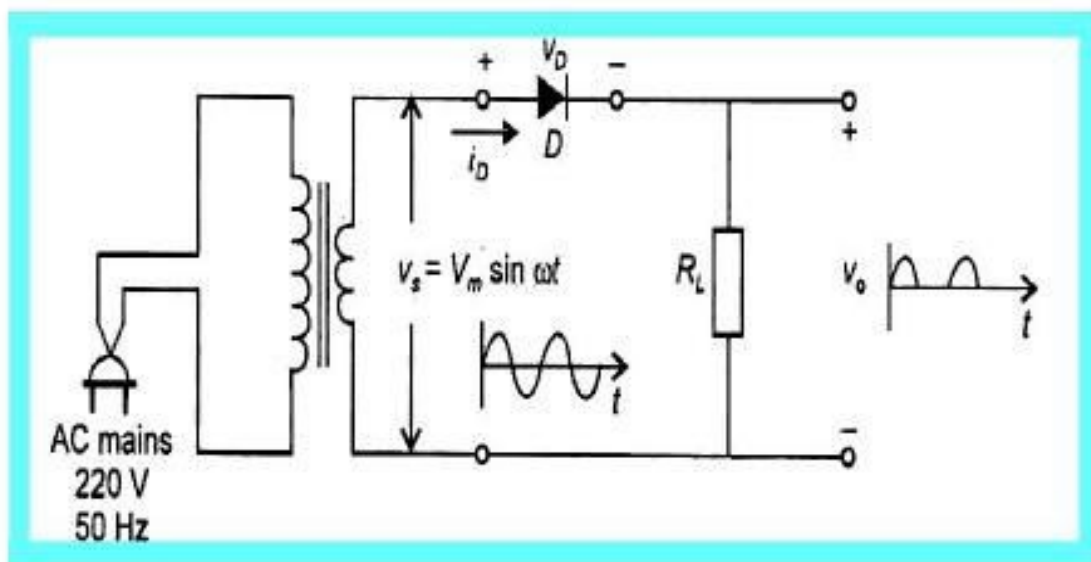


Figure 1: Half wave rectifier

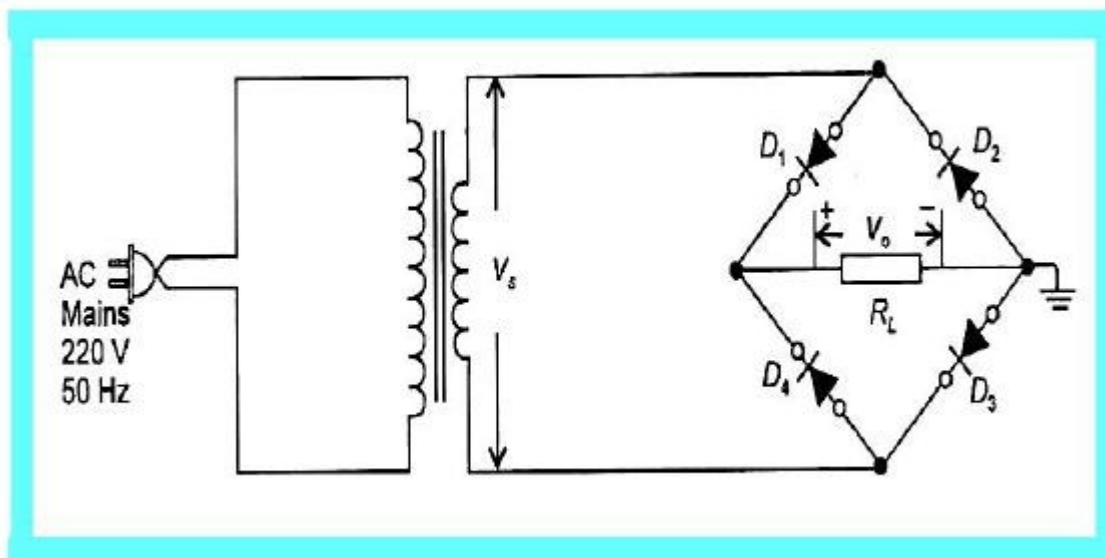


Figure 2 : Full wave bridge rectifier

PROCEDURE:

1. Connect the circuit as shown in figure.
2. Apply A.C. supply through transformer;
3. Find the current through load resistance.
3. Observed voltage wave across the load on CRO.
4. Find the value of ripple factor and efficiency.

Calculation for Half Wave Rectifier:

Ripple factor

$$r = \frac{\text{rms value of the ac components of wave}}{\text{average or dc value}}$$

$$r = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1} = \sqrt{\left(\frac{I_m/2}{I_m/\pi}\right)^2 - 1} = 1.21$$

Efficiency of half wave rectifier

Efficiency, is the ratio of the dc output power to ac input power

Calculation for Full Wave Bridge Rectifier:

$$\eta = \frac{P_{dc}}{P_{ac}} = \frac{I_{dc}^2 R_L}{I_{rms}^2 (R_L + r_f)} = \left(\frac{I_{dc}}{I_{rms}}\right)^2 \frac{R_L}{R_L + r_f}$$
$$= \left(\frac{I_m/\pi}{I_m/2}\right)^2 \frac{R_L}{R_L + r_f} = 0.406 \frac{R_L}{R_L + r_f}$$

$$= \frac{40.6}{1 + r_f/R_L} \%$$

Ripple Factor

The ripple factor for a Full Wave Rectifier is given by

$$r = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1} = \sqrt{\left(\frac{I_m/\sqrt{2}}{2I_m/\pi}\right)^2 - 1} = \sqrt{\left(\frac{\pi}{2\sqrt{2}}\right)^2 - 1}$$

Efficiency

Efficiency, is the ratio of the dc output power to ac input power.

RESULTS:

Voltage wave form across the load:

Ripple factor:

Efficiency:

LEARNING OUTCOMES:

EXPERIMENT NO. : 6

AIM:

To study the forward and reverse bias volt-ampere characteristics of a zener diode. Also determine the breakdown voltage, static and dynamic resistances.

APPARATUS AND COMPONENTS REQUIRED:

Sl. No	Apparatus	Specifications	Quantity
1	DC Power Supply	-	1
2	Bread Board	-	1
3	Multimeter	-	1
4	Connecting wire	-	-
5	Zener Diode	$V_z=5.1V$	1
6	Resistors	$1K\Omega$	1

THEORY:

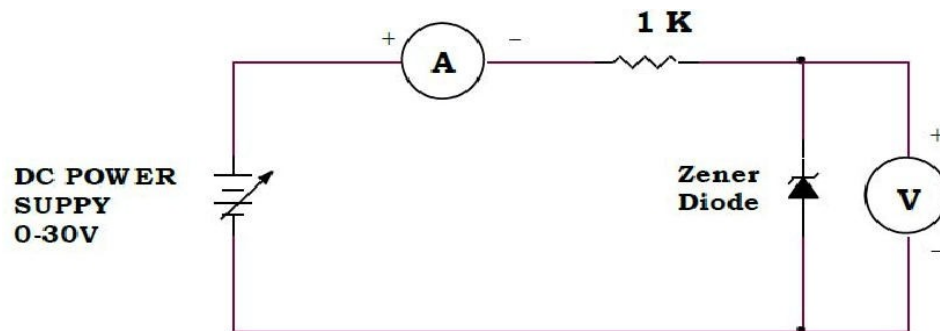
The circuit diagram to plot the VI characteristics of a *zener* diode is shown. Zener diode is a special diode with increased amounts of doping. This is to compensate for the damage that occurs in the case of a *pn* junction diode when the reverse bias exceeds the breakdown voltage and thereby current increases at a rapid rate.

Applying a positive potential to the anode and a negative potential to the cathode of the zener diode establishes a forward bias condition. The forward characteristic of the zener diode is same as that of a *pn* junction diode i.e. as the applied potential increases the current increases exponentially. Applying a negative potential to the anode and positive potential to the cathode reverse biases the zener diode.

As the reverse bias increases the current increases rapidly in a direction opposite to that of the positive voltage region. Thus under reverse bias condition breakdown occurs. It occurs because there is a strong electric field in the region of the junction that can disrupt the bonding forces within the atom and generate carriers. The breakdown voltage depends upon the amount of doping. For a heavily doped diode depletion layer will be thin and breakdown occurs at low reverse voltage and the breakdown voltage is sharp. Whereas a lightly doped diode has a higher breakdown voltage. This explains the zener diode characteristics in the reverse bias region. The maximum reverse bias potential that can be applied before entering the zener region is called the Peak Inverse Voltage referred to as PIV rating or the Peak Reverse Voltage Rating (PRV rating)

CIRCUIT DIAGRAM

(REVERSE BIAS):



(FORWARD BIAS):

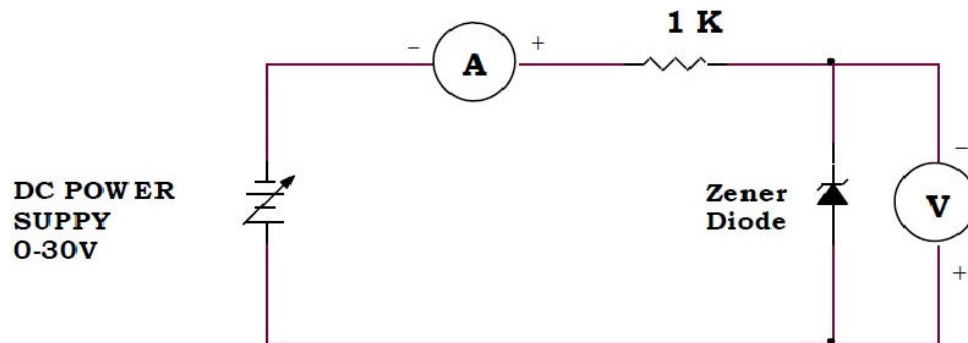


Figure 1 : Circuit diagram for Reverse and Forward bias characteristics of Zener Diode

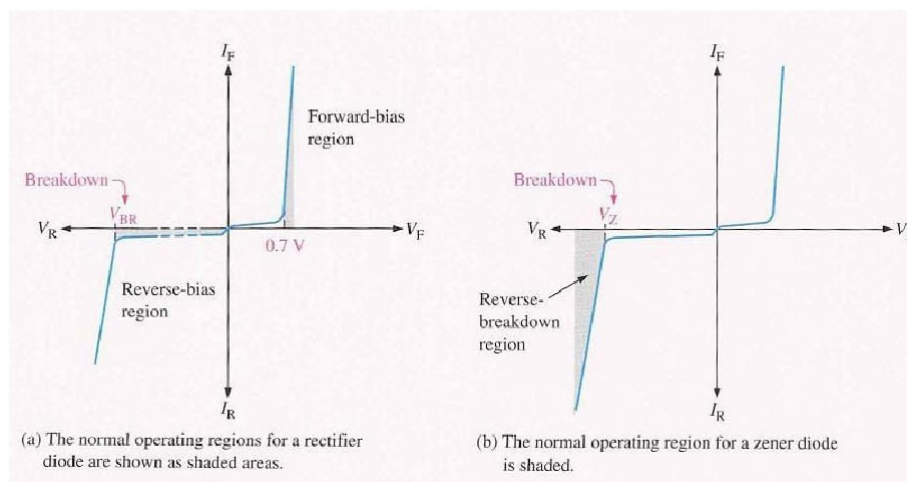


Figure 2: I-V plot of Zener Diode

PROCEDURE:

1. Connect the circuit as shown in figure.
2. Apply DC supply as shown in the figure.
3. Measure the value of V_Z and I_Z , V_R Reverse biased.
4. Repeat the all step with opposite polarity of zener diode.
5. Measure the value of V_Z and I_Z , V_F Reverse biased.
6. Plot the curve between Voltage and current for forward and reverse biased zener diode.

OBSERVATION:

S. No.	Forward Bias				Reverse Bias			
	V_F (V)	V_Z (V)	V_r (V)	$I_Z = V_r/R$ (mA)	V_R (V)	V_z (V)	V_r (V)	$I_z = V_r/R$ (mA)
1.								
2.								
3.								

RESULT:

Plot for reverse biased and forward biased zener diode: Break down voltage value:

Static resistance = V_Z/I_Z

Dynamic resistance value: change in zener voltage/change in current through zener diode.

LEARNING OUTCOMES:

EXPERIMENT NO.: 7

AIM:

To plot input characteristic of a BJT in Common Emitter configuration.

APPARATUS AND COMPONENTS REQUIRED:

Sl. No	Apparatus	Specifications	Quantity
1	Digital Multimeter	-	1
2	Bread Board	-	1
3	DC Power Supply	-	1
4	Connecting wire	-	-
5	Transistor	BC547	1
6	Resistors	1K Ω , 100K Ω	1each

THEORY:

A transistor is a three terminal device. The terminals are emitter, base, collector. In common emitter configuration, input voltage is applied between base and emitter terminals and output is taken across the collector and emitter terminals. Therefore the emitter terminal is common to both input and output ports.

The input characteristic resembles that of a forward biased diode curve. This is expected since the Base-Emitter junction of the transistor is forward biased. As compared to CB arrangement I_B increases less rapidly with V_{BE} . Therefore input resistance of CE circuit is higher than that of CB circuit.

The current amplification factor of CE configuration is given by

$$\beta = \Delta I_C / \Delta I_B$$

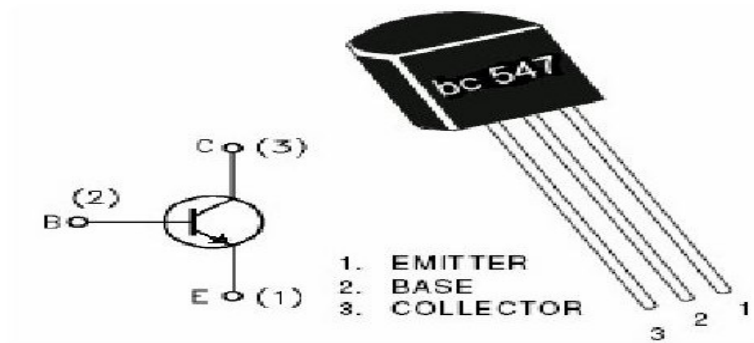


Figure 1: Transistor Symbol

INPUT CHARACTERISTIC

- 1) Connect the circuit as per the circuit diagram
- 2) For plotting the input characteristics the output voltage V_{CE} is kept constant at 1V and by varying $V_{BB}(0-0.1-2V)$ note down different values of V_{BE} , V_{RB} and I_B
- 3) Repeat the above step by keeping V_{CE} at 2V and 3V.
- 4) Tabulate all the readings.
- 5) Plot the graph between V_{BE} and I_B for constant V_{CE} .

[illegible]

INPUT CHARACTERISTIC GRAPH

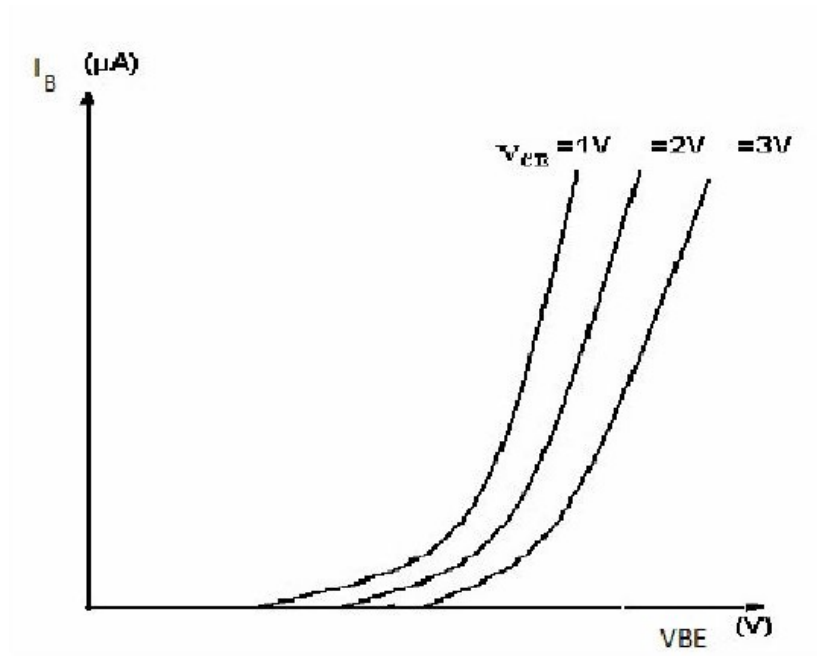


Figure 3: Input characteristics of the NPN transistor in common emitter configuration

RESULT:

LEARNING OUTCOME:

EXPERIMENT NO.: 8

AIM:

To plot output characteristic of a BJT in Common Emitter Configuration.

APPARATUS AND COMPONENTS REQUIRED:

Sl. No	Apparatus	Specifications	Quantity
1	Digital Multimeter	-	1
2	Bread Board	-	1
3	DC Power Supply	-	1
4	Connecting wire	-	-
5	Transistor	BC547	1
6	Resistors	1K Ω , 100K Ω	1each

THEORY:

A transistor is a three terminal device. The terminals are emitter, base, collector. In common emitter configuration, input voltage is applied between base and emitter terminals and output is taken across the collector and emitter terminals. Therefore the emitter terminal is common to both input and output

The output characteristics are drawn between I_c and V_{CE} at constant I_B . the collector current I_c varies with V_{CE} upto few voltage only. After this the collector current becomes almost constant and independent of V_{CE} . The value of V_{CE} upto which the collector current changes with V_{CE} is known as knee voltage. The transistor always operates in the region above knee voltage. I_c is always constant and is approximately equal to I_B .

The current amplification factor of CE configuration is given by

$$\beta = \Delta I_C / \Delta I_B$$

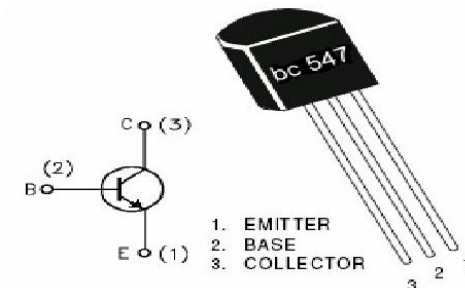


Figure 1: Transistor Symbol

OUTPUT CHARACTERISTIC

- 1) Connect the circuit as per the circuit diagram.
- 2) For plotting the output characteristics the input current I_B is kept constant at $10\mu A$ and by varying V_{CC} note down the different values of V_{CE} , V_{RB} and I_C .
- 3) Repeat the above step by keeping I_B at $20\mu A$ $40\mu A$.

S. No.	$I_{B1}=10\mu A$				$I_{B2}=20\mu A$				$I_{B3}=40\mu A$			
	V_{CC} (mv)	V_{CE} (mv)	V_{RC} (mv)	I_C (mA)	V_{CC} (mv)	V_{CE} (mv)	V_{RC} (mv)	I_C (mA)	V_{CC} (mv)	V_{CE} (mv)	V_{RC} (mv)	I_C (mA)
1.												
2.												
3.												
4.												
5.												

OUTPUT CHARACTERISTIC GRAPH

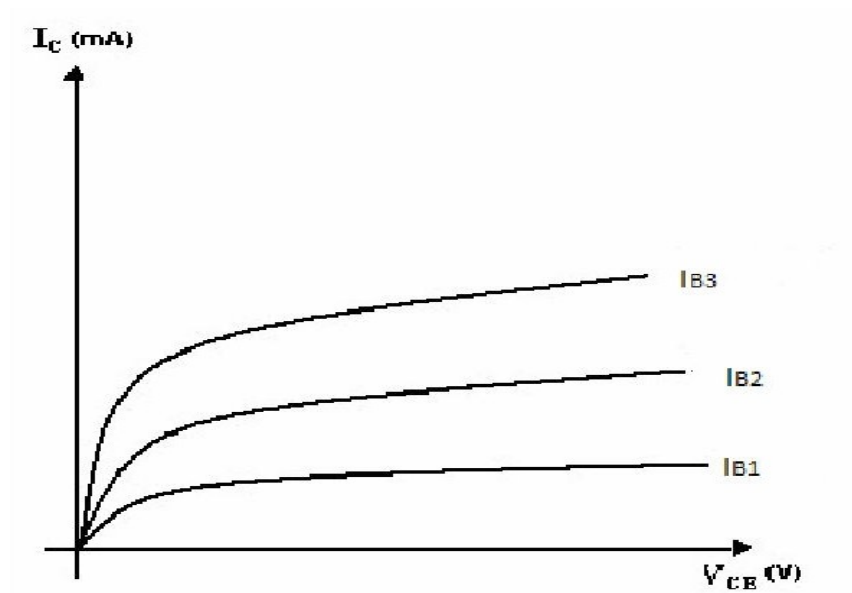


Figure 3: Output characteristics of the NPN transistor in common emitter configuration

RESULT:

LEARNING OUTCOME:

EXPERIMENT NO.: 9

AIM:

To observe inverting and non-inverting amplifier configurations using Op-Amp IC741.

APPARATUS AND COMPONENTS REQUIRED:

Sl.No	Apparatus	Specifications	Quantity
1	CRO	-	1
2	Power Supply	-	1
3	Bread Board	-	1
4	Connecting wire	-	-
5	Op Amp	IC -741	1
6	Resistors	1K Ω , 10K Ω	2 each

THEORY:

Pin Diagram of Op-Amp IC741:

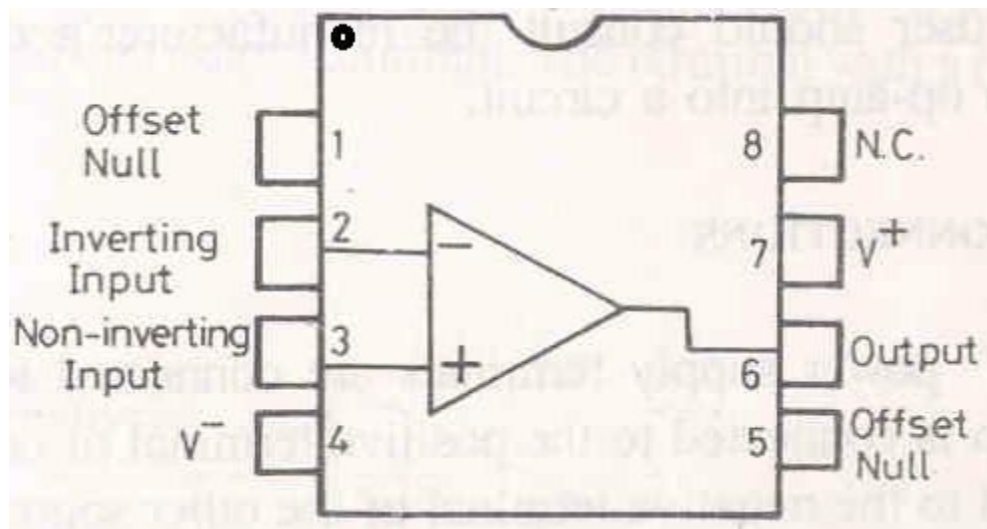


Figure 1 : Pin Diagram of Op-Amp741 IC

Inverting amplifier

Figure 2(a) shows an inverting amplifier. The output voltage (V_o) is of the opposite polarity as the input voltage (V_{in}). The input signal is applied directly to the inverting (-ve) input terminal of the amplifier and the feedback resistance is also connected between the output terminal and inverting input terminal.

Closed –Loop Voltage Gain (A_F):

The closed-loop voltage gain A_F can be obtained by writing Kirchhoff's Current equation at the input node 2 as follows:

$$I_{in} = I_F + I_B \dots\dots\dots (1)$$

Since R_i is very large, the input bias current (I_B) is negligibly small. For

instance, $R_i = 2M\Omega$ and $I_B = 0.5\mu A$ for 741C. Therefore, $I_{in} \cong I_F$

Now, from circuit diagram shown in Fig. 2 (a)

$$\frac{V_{in} - V_2}{R_1} = \frac{V_2 - V_o}{R_F} \dots\dots\dots (2)$$

Using Virtual short concept $V_1 = V_2$

Since $V_2 = 0V$, Substituting this value of V_2 in Equation (2)

$$A_F = \frac{V_o}{V_{in}} = \frac{-R_F}{R_1} \dots\dots\dots (3)$$

CIRCUIT DAIGRAM & WAVEFORM

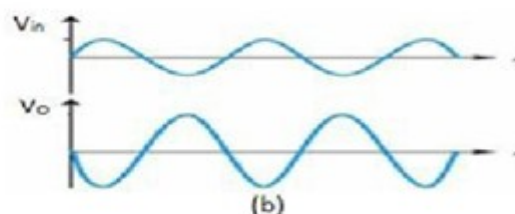
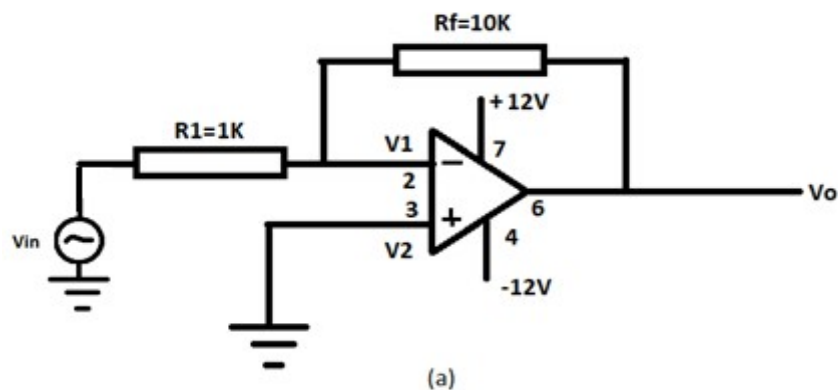


Figure 2: Inverting Amplifier (a) Circuit diagram (b) Waveforms

PROCEDURE:

1. Make the connections as per the circuit diagram.
2. Apply the input to the Inverting terminal and obtain the output on the CRO.
3. Take various readings by varying the input voltage and hence calculate the gain.
4. Calculate the gain using formula.

$$A_F = - R_F / R_1 \text{ (Theoretical)}$$

$$A_F = V_o / V_{in} \text{ (Practical)}$$

OBSERVATION TABLE:

S.No.	V _{in} (volts)	V _o (volts)	A _F = - R _F /R ₁ (Theoretical)	A _F = V _O /V _{in} (Practical)
1				
2				
3				
4				

Non inverting amplifier

Figure shows a non-inverting amplifier. The output Voltage (V_o) is of the same polarity as the input voltage (V_{in}.) The input resistance of the non-inverting amplifier is very large (100MΩ) in this case. The input signal is applied directly to the non-inverting (+ve) input terminal of the amplifier and the feedback resistance are connected between the output terminal, the –ve input terminal and ground.

The Minimum Gain of the Non-Inverting Amplifier is 1

Closed –Loop Voltage Gain (A_F):

$$A_F = V_o / V_{in} \dots\dots\dots (4)$$

From the circuit diagram shown in Fig. 3 (a)

$$V_1 / R_1 = (V_o - V_1) / R_F \dots\dots\dots (5)$$

Using Virtual short concept V₁=V₂=V_{in}

$$\text{Therefore, } V_{in} / R_1 = (V_o - V_{in}) / R_F$$

$$A_F = V_o / V_{in} = 1 + R_F / R_1 \dots\dots\dots (6)$$

CIRCUIT DAIGRAM & WAVEFORM

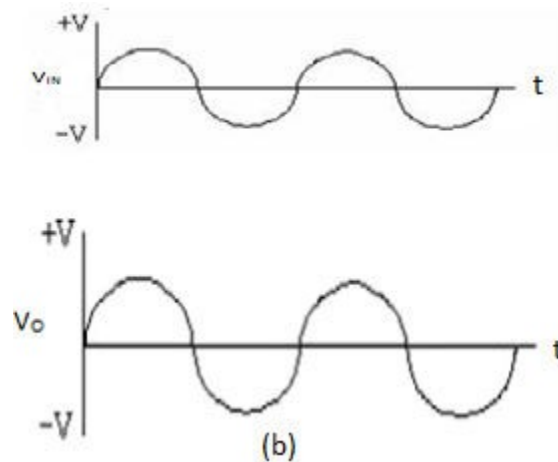
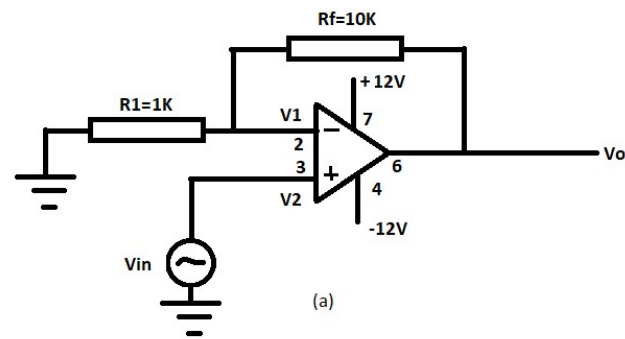


Figure 3: Non-Inverting Amplifier (a) Circuit diagram (b) Waveforms

PROCEDURE:

1. Make the connections as per the circuit diagram.
2. Apply the input to the Non-Inverting terminal and obtain the output on the CRO.
3. Take various readings by varying the input voltage and hence calculate the gain.
4. Calculate the gain using formula.

$$A_F = 1 + R_F / R_1$$

$$A_F = V_o / V_{in}$$

OBSERVATION TABLE:

S. No.	V_{in} (Volts)	V_o (Volts)	$A_F=1+(R_F/R_1)$ (Theoretical)	$A_F=V_o/V_{in}$ (Practical)

RESULT:

Voltage gain of the Inverting/Non-inverting amplifiers depends on the value of R_F & R_I , whereas it is independent of the open loop gain ' A_o ' of the op-amp. Theoretical and Practical values of the Gain are approximately same.

LEARNING OUTCOME:

EXPERIMENT NO : 10

AIM:

To realize an adder and subtractor circuits using Op- Amp IC 741 amplifier.

APPARATUS AND COMPONENTS REQUIRED:

Sl.No	Apparatus	Specifications	Quantity
1	DC Regulated Power Supply	+/- 12 V Dual Supply, 0-5V, 0-10V	01
2	Digital Multimeter	-	1
3	Bread Board	-	1
4	Connecting wire	-	-
5	Op Amp	IC -741	1
6	Resistors	1k Ω	4
7	Resistors	10k Ω	1

THEORY:

ADDER:

Op-amp can be used to design a circuit whose output is the sum of several input signals. Such a circuit is called a summing amplifier or an adder. Summing amplifier can be classified as inverting and non-inverting summer depending on the inputs applied to inverting and non-inverting terminals respectively. Figure 1 shows a non-inverting adder with n inputs. Here the output will be the linear summation of input voltages. The circuit can also be used as summing amplifier, scaling amplifier, or as an averaging amplifier also.

From the circuit of adder, it can be noted that at pin3, $I_1 + I_2 + I_3 + \dots \dots \dots n = 0$

$$\begin{aligned} \frac{V_a - V_1}{R} + \frac{V_a - V_2}{R} + \frac{V_a - V_3}{R} \dots \dots \dots \frac{V_a - V_n}{R} &= 0 \\ nV_a - \frac{(V_1 + V_2 + V_3 + \dots \dots \dots + V_n)}{R} &= 0 \\ V_a = \frac{V_1 + V_2 + V_3 + \dots \dots \dots + V_n}{R} &= 0 \\ V_o = \left(1 + \frac{R_f}{R_1} \right) V_a \\ V_o = \left(1 + \frac{R_f}{R_1} \right) \left(\frac{V_1 + V_2 + V_3 + \dots \dots \dots + V_n}{n} \right) \\ = 1 + (n-1) \left(\frac{V_1 + V_2 + V_3 + \dots \dots \dots + V_n}{n} \right) \\ = n \left(\frac{V_1 + V_2 + V_3 + \dots \dots \dots + V_n}{n} \right) \\ V_o = V_1 + V_2 + V_3 + \dots \dots \dots + V_n \end{aligned}$$

The output voltage should be equal to the sum of all the input voltages.

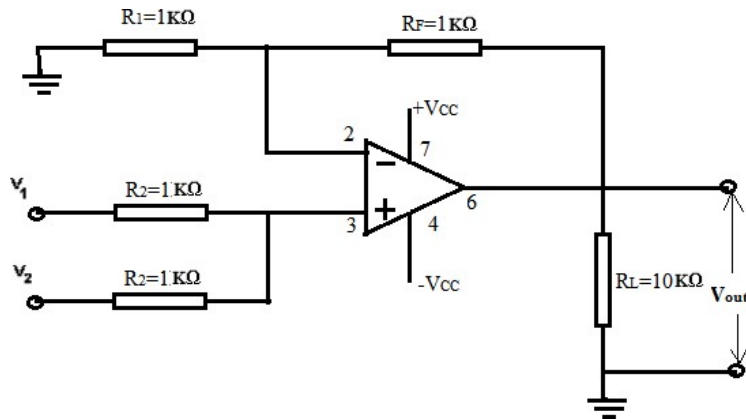


Figure 1: Circuit diagram of Op-Amp as Adder

PROCEDURE:

1. Connect the adder circuit as shown in Figure1.
2. Apply the input voltages V_1 , V_2 from DC power supply (Negative of the power supply should be connected with the ground).
3. Biasing of the Op-amp should be done with correct polarity of V_{cc} (+/-12V).
4. Measure the output voltage across R_L with the help of DMM.

OBSERVATION TABLE:

	V_1 (V)	V_2 (V)	V_{out} (V)	$V_1 + V_2$ (V)
Theoretical values				
Measured Values				

SUBTRACTOR:

A subtractor is a circuit that gives the difference of the two inputs, $V_o = V_2 - V_1$, where V_1 and V_2 are the inputs. By connecting one input voltage V_1 to inverting terminal and another input voltage V_2 to the non-inverting terminal, we get the resulting circuit as the Subtractor. This is also called as differential or difference amplifier using op-amps.

Output of a differential amplifier (subtractor) is given as

$$V_o = (-R_f/R_1) (V_1 - V_2)$$

If all external resistors are equal in value, then the gain of the amplifier is equal to -1. The output

voltage of the differential amplifier with a gain of -1 is

$$V_o = (V_2 - V_1)$$

Thus the output voltage V_o is equal to the voltage V_2 applied to the non – inverting terminal minus the voltage V_1 applied to the inverting terminal.

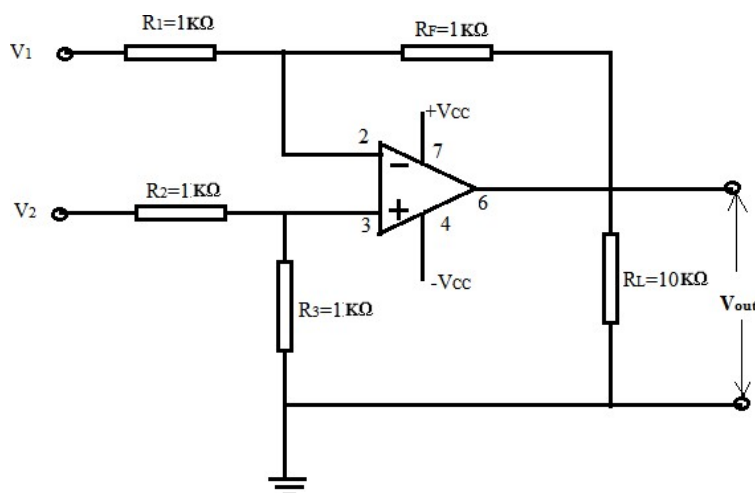


Figure 2: Circuit diagram of Op-Amp as Subtractor

PROCEDURE:

1. Connect the subtractor circuit as shown in Figure 2.
2. Apply the input voltages V_1 , V_2 from DC power supply (Negative of the power supply should be connected with the ground).
3. Biasing of the Op-amp should be done with correct polarity of V_{cc} (+/-12V).
4. Measure the output voltage across R_L with the help of DMM.

OBSERVATION TABLE:

	V_1 (V)	V_2 (V)	V_{out} (V)	$V_2 - V_1$ (V)
Theoretical values				
Measured Values				

RESULT:

LEARNING OUTCOMES:

EXPERIMENT NO : 11

AIM:

Verification of Superposition Theorem.

APPARATUS AND COMPONENTS REQUIRED:

Sl.No	Apparatus	Specifications	Quantity
1	Multimeter	Digital	1
2	DC Regulated Power Supply	+5V, +12V	1
3	Bread Board	--	1
5	Connecting wire	-	
5	Resistors	1 K Ω , 4.7 K Ω , 2.2 K Ω	1 each

THEORY:

In any linear bilateral network containing two or more independent sources (voltage or current sources or combination of voltage and current sources) the resultant current / voltage in any branch is the algebraic sum of currents / voltages caused by each independent sources acting alone, with all other independent sources being replaced meanwhile by their respective internal resistances.

CIRCUIT DIAGRAM:

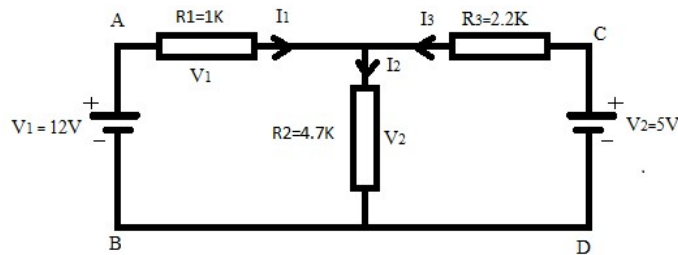


Figure 1 Circuit diagram

PROCEDURE:

1. Measure the practical value of the given resistance with the help of Digital multimeter.
2. Make the circuit on the breadboard as per Figure 2,

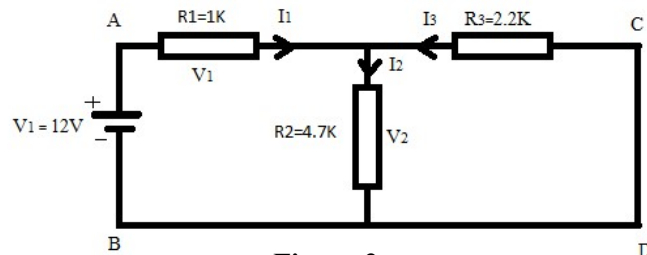


Figure 2

4. Consider source V_1 and open the DC source V_2 , replace it by a short circuit (internal resistance of source) by wire. Measure voltage across V_{R1} , V_{R2} and V_{R3}
5. Now consider source V_2 and open the DC source V_1 , replace it by a short circuit (internal resistance of the source). Measure voltage across R_1 , R_2 and R_3 Figure 3.

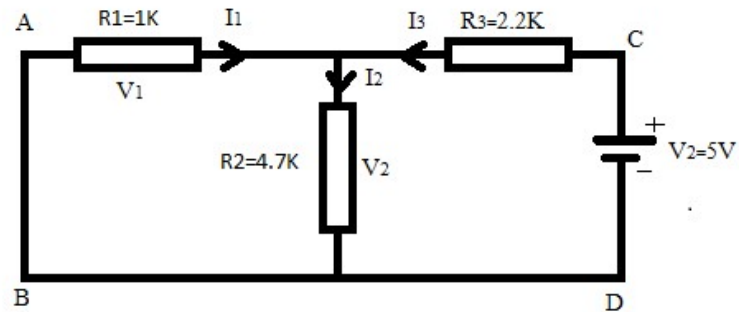


Figure 3

Note :- Keep the polarity of the DMM lead same while taking measurement.

6. Now consider both the sources V_1 and V_2 . Measure voltage across R_1, R_2 and R_3 . (Figure:1)

OBSERVATION TABLE:

Table I – Resistance Measurements

Value Determination	R_1	R_2	R_3
Color Code			
Measured			

Table II – Voltage Supply Measurements

Value Determination	V_1	V_2
Nominal	12 V	5V
Measured		

Table III – MEASURED Currents and Potentials

Active V	V_{R_1}	V_{R_2}	V_{R_3}	I_1	I_2	I_3
V_1 ONLY						
V_2 ONLY						
BOTH						
Calc $\Delta\%$						

$\Delta\% = 100 \cdot [X_{\text{BOTH}} - (\sum X_j)] / X_{\text{BOTH}}$ Where $X_j \equiv$ either of V_{R_j} or I_j

Table IV – CALCULATED Currents and Potentials

Active V	VR1	VR2	VR3	I ₁	I ₂	I ₃
V ₁ ONLY						
V ₂ ONLY						
BOTH						

RESULT:

PRECAUTION:

LEARNING OUTCOME:

EXPERIMENT NO: 12

AIM:

Realization of desired wave shapes using clipper and clamper circuits.

APPARATUS AND COMPONENTS REQUIRED:

Sl.No	Apparatus	Specifications	Quantity
1	CRO	-	1
2	Power Supply	DC regulated	1
3	Function Generator		1
4	Bread Board	--	1
5	Connecting wire	-	
6	Resistors	1 K Ω , 10 K Ω , 100 K Ω	1each
7	Capacitor	0.1 μ F	1
8	Diode	IN4007	1

THEORY:

The circuit with which the waveform is shaped by removing (or clipping) a portion of the input signal without distorting the remaining part of the alternating waveform is called a *clipper*. Clipping circuits are also referred to as voltage (or current) limiters, amplitude selectors, or slicers. These circuits find extensive use in radars, digital computers, radio and television receivers etc.

In a clipping circuit, the output voltage will be proportional to the input voltage as long as the input lies between the specified reference levels. Outside this range, the output is clipped - it remains essentially constant, no longer dependent on the input. Clipping circuits find important uses in wave shaping and signal processing applications.

Often in the development of electronic circuits it is required that voltages be limited in some manner to avoid circuit damage. Furthermore, the limiting or clipping of voltages can be very useful in the development of wave-shaping circuits. A typical clipper circuit is shown in Figure 1. In this circuit the output voltage can never be greater than 3 V. The ideal diode becomes forward biased at V_o equal to 3 V and this ties the output directly to the 3 V supply. The waveform can be

clipped on the negative side by placing the series combination of a diode and power supply in parallel with the diode and power supply already shown.

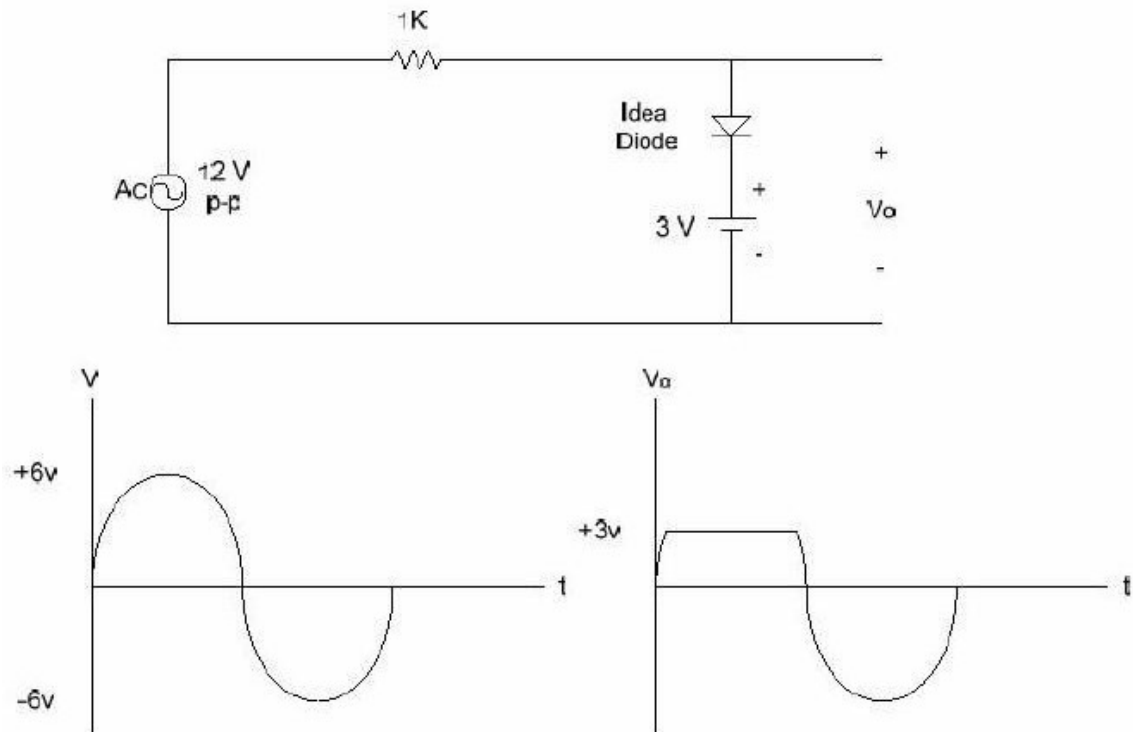


Figure 1: Diode clipping circuit showing input and output waveforms

While clipper circuits are concerned primarily with limiting or cutting off part of the waveform, clampers are used primarily to shift the DC level. For example, if we have a clock signal that swings between 0v and 5V but our application requires a clock signal from -5V to 0V, we can provide the proper DC offset with a passive clamper circuit. A typical clamper circuit is shown in Figure 2. For this circuit to work properly the pulse width needs to be much less than the RC timeconstant of 10 ms. The input square wave with a frequency of 1 KHz and a pulse width of 0.5 ms meets this requirement. The diode and power supply as shown will prevent the output voltage from exceeding 3 V (i.e., all of the region above 3 V can be viewed as a forbidden region for output voltage). Because of the time constant requirement the voltage across the capacitor cannot change significantly during the pulse width, and after a short transient period the voltage across the capacitor reached a steady state offset value. The output voltage is simply the input voltage shifted by this steady state offset. Also, observe that the peak-to-peak output voltage is equal to the peak-to-peak input voltage. This is true because the voltage across the capacitor cannot change instantaneously and the full change of voltage on the input side of the capacitor will likewise be seen on the output side of the capacitor.

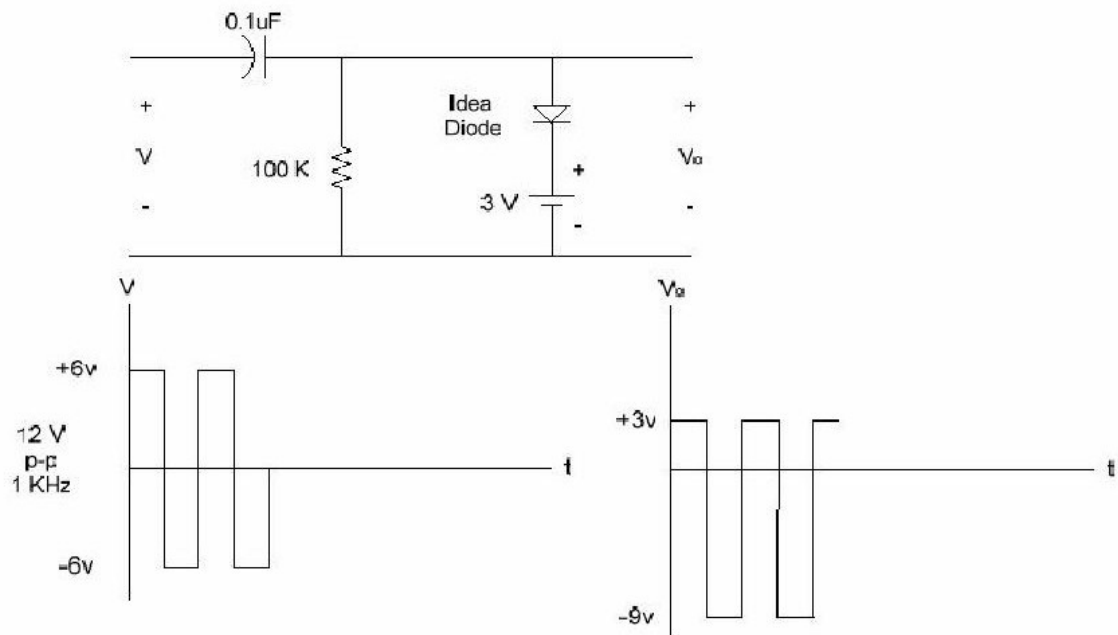
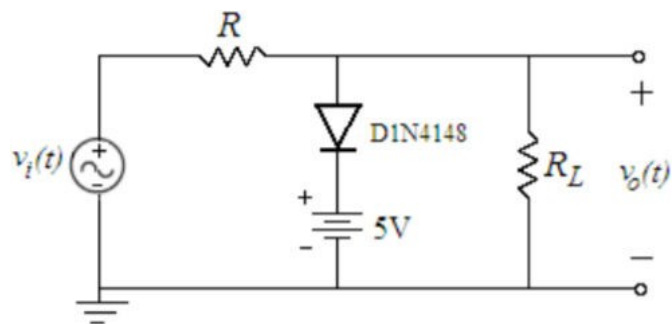


Figure 2: Diode clamper circuit showing input and output waveforms

PROCEDURE:

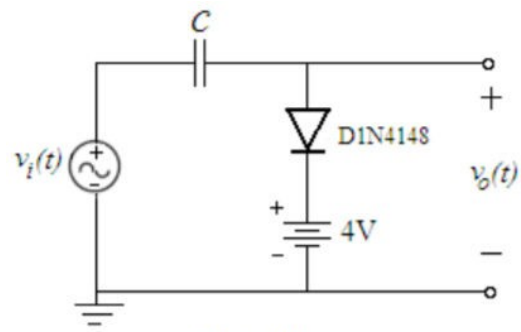
A. Clipper Circuits:

1. Consider the clipper circuit in Fig.3, draw the input and output voltage waveforms in same plot
 $v_i(t) = 10\sin(200\pi t)$, $R = 1\text{ k}\Omega$ and $R_L = 47\text{ k}\Omega$.



figure_3

2. Consider the clamper circuit in Fig.4, draw the input and output voltage waveforms, assume
 $v_i(t) = 10\sin(200\pi t)$ and $C = 47\mu\text{F}$.



figure_4

RESULT:

PRECAUTION:

LEARNING OUTCOME: